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A PROJECT REPORT ON
IDENTIFICATION OF MEDICINAL PLANTS USING IMAGE
PROCESSING WITH DEEP LEARNING TECHNIQUES

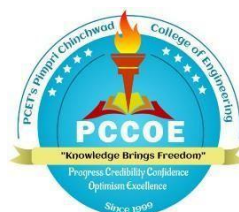
SUBMITTED TO THE PIMPRI CHINCHWAD COLLEGE OF
ENGINEERING AN AUTONOMOUS INSTITUTE, PUNE
IN THE FULFILLMENT OF THE
REQUIREMENTS FOR THE AWARD OF
THE DEGREE
OF

BACHELOR OF TECHNOLOGY
COMPUTER ENGINEERING (REGIONAL LANGUAGE)

SUBMITTED BY

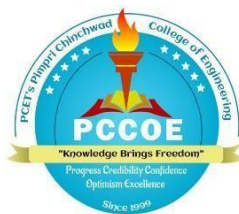
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2024-25



CERTIFICATE

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“**IDENTIFICATION OF MEDICINAL PLANTS USING IMAGE PROCESSING WITH
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ABSTRACT

The identification and authentication of medicinal plants hold significant importance in domains such as healthcare, biodiversity conservation, pharmaceutical research, and traditional medicine systems like Ayurveda and Unani. Conventionally, plant identification relies heavily on the expertise of trained botanists who analyze morphological traits such as leaf shape, flower structure, and plant habitat. However, these manual methods are often time-consuming, subjective, and prone to human error particularly when dealing with a vast number of plant species or incomplete specimens.

With the advent of technological advancements in computer vision and artificial intelligence, there is a growing shift toward automated and scalable solutions for biological classification. This project proposes a novel methodology for the identification and classification of medicinal plants using leaf images, leveraging Convolutional Neural Networks (CNNs)—a class of deep learning algorithms known for their exceptional performance in image recognition tasks.

The core idea revolves around analyzing distinct leaf features such as shape, venation, margin, and texture to accurately identify plant species. A comprehensive dataset of medicinal plant leaf images was collected, pre-processed, and used to train a CNN-based model capable of distinguishing among multiple species with high precision. The model's architecture was optimized through hyperparameter tuning and performance evaluation metrics such as accuracy, precision, recall, and F1-score.

The outcomes of the experimental analysis demonstrate a significant improvement in classification accuracy compared to traditional machine learning approaches, showcasing the effectiveness of CNNs in this application. The proposed system offers a fast, reliable, and user-friendly platform for medicinal plant identification, which can serve as a practical tool for botanists, healthcare practitioners, pharmaceutical researchers, and students.

By automating the process of plant identification, this research contributes to preserving traditional knowledge, enhancing accessibility to medicinal resources, and supporting sustainable practices in herbal medicine. Moreover, it lays a solid foundation for further research involving real-time detection, mobile deployment, and integration with larger botanical databases.

KEYWORDS - Medicinal Plants, Plant Identification, Healthcare, Image Processing, Machine Learning, Leaf Classification, Plant Species, Leaf Features, Convolutional Neural Network (CNN), Deep Learning, Computer Vision, Artificial Intelligence.

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LIST OF ABBREVIATIONS

Abbreviation	Illustration
CNN	Convolutional Neural Networks
IMP	Identification of Medicinal Plants
SVM	Support Vector Machine
IP	Image Processing
ML	Machine Learning
VT	Voice Technology
DL	Deep Learning
MPVT	Medicinal Plant Voice
VGG	Visual Geometry Group

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Chapter 1

Introduction

1.1 Domain Description

The domain identification of the medicinal plant intersects several critical areas including health care, protection of biodiversity, pharmacognosy and ethnobotany. Medicinal plants form the cornerstone of traditional medical systems and are increasingly recognized by modern pharmacological science for their therapeutic potential. However, the exact identification and verification of these plants remains a significant challenge, especially in field conditions and rural or remote areas where the expertise is rare.

Traditionally, the process of identifying medicinal plants relied on manual methods involving visual control of morphological characteristics such as the shape of the leaves, the formulas, flowers, the texture of the bark and other plant structures. These methods require considerable botanical expertise and can be time -consuming, subjective and prone to irregularities. The reliance on physical samples and professional taxonists also reduce the scalability and availability of these identification processes, especially for communities or individuals without access to trained experts.

With continuing technological development, especially in the field of image processing, artificial intelligence (AI) and machine learning (ML), an innovative solution has appeared to automate and simplify the process of plant identification. In this context, the classification of leaves based on plants has gained significant traction due to the widespread availability of digital imaging devices and abundance of leaves of leaves during the life cycle of the plant. The leaves offer a consistent and accessible set of morphological elements, which makes them ideal for classification tasks using computer vision techniques.

This research focuses on the development of an identification system based on machine learning, which uses a convolutional neural network (CNN) for automated classification of medicinal plant species using images of their leaves. CNNs, a specialized class of deep learning models designed for image -related tasks, is able to automatically learn hierarchical properties from raw pixels, which is particularly suitable for complex patterns recognition tasks such as classification of leaves.

The proposed system captures and analyzes various distinguishing features of plant leaves - including texture, shape, edge contours and vein structures - allowing precise categorization of plant species. This approach eliminates the need for manual extraction of the functions and intervention of experts, thereby significantly shortening the time and effort required to identify plants.

Integration of CNN -based models with curatorial data set of medicinal plants is to create a scalable, effective and accurate system that can help scientists, students, herbal doctors and nature in identifying

medicinal plants in real time. Furthermore, this methodology has the potential for integration into mobile and web platforms and extends its availability and practical usability in the real-world scenarios.

In short, this research examines the convergence of deep learning and botanical science to solve the long -term call of identification of medicinal plants. Through the CNNS study, the study demonstrates the ability of machine learning models to replicate and in many cases increases the accuracy of traditional identification techniques - which will strengthen the path for future developments in automated plant classification systems.

1.2 Problem Definition

The accurate identification and authentication of medicinal plants play a vital role in supporting global healthcare, biodiversity conservation, and the preservation of traditional medicinal practices. Traditional methods of plant identification rely heavily on manual analysis by expert botanists, which is time-consuming, error-prone, and often inaccessible—particularly in rural and resource-limited areas.

With the advancement of artificial intelligence, particularly in the domains of image processing and deep learning, there exists a promising opportunity to automate and accelerate the plant identification process. However, challenges persist in building systems that are not only accurate and reliable but also scalable and adaptable across a wide range of species.

This project aims to develop and implement a robust, intelligent, and efficient system for the automated detection and classification of medicinal plant species based on the analysis of leaf images. The proposed solution leverages Convolutional Neural Networks (CNNs), with a specific implementation of the VGG19 deep learning architecture, to perform high-precision classification by learning complex leaf features such as shape, texture, venation, and edge patterns.

The problem being addressed is twofold:

1. **To replace traditional plant identification methods** with a faster, scalable, and automated approach using image-based recognition.
2. **To improve classification accuracy** by applying state-of-the-art deep learning techniques—specifically the pre-trained VGG19 model—suitable for real-world deployment in mobile or web applications.

1.3 Goals and Objectives

The primary goal of this project is to design and develop a deep learning-based system that can accurately identify medicinal plants through leaf image analysis. The project focuses on leveraging advanced Convolutional Neural Network (CNN) architectures, with an emphasis on improving the accuracy, efficiency, and scalability of existing plant identification systems.

The key objectives of the project are as follows:

- To explore and implement deep learning techniques using well-established architectures such as VGG16, MobileNet, DenseNet, and other CNN-based models for plant classification through image processing.
- To develop a robust classification model that can accurately detect and differentiate between various species of medicinal plants using visual features extracted from leaf images.
- To perform training, testing, and validation of the proposed models using a curated dataset of medicinal plant leaf images, and to fine-tune the models for optimal performance.
- To evaluate the performance of different CNN architectures based on key metrics such as accuracy, precision, recall, and F1-score, and determine the most effective model for real-world applications.
- To enhance the existing plant identification systems by introducing automation, scalability, and improved classification capabilities using state-of-the-art deep learning algorithms.

1.4 Motivation

Medicinal plants have been an integral part of traditional health systems for centuries and continues to serve as a primary source of inspiration for the development of modern pharmaceutical drugs. Many life -saving lives used today are derived from compounds originally found in medicinal plants. However, the exact identification of these plants remains a significant challenge, especially in distant regions, where access to professional botany and diagnostic tools is limited.

The motivation of this project stems from the urgent need to maintain traditional knowledge, speed up the discovery of plant medicinal products and use modern technologies to make health care more accessible and efficient. By using deep learning techniques - specifically a convolutional neural network (CNN) - we try to create a bridge between ancient herbal knowledge and modern artificial intelligence.

The key motivational factors include:

- **Increasing health care and discovery of drugs:** precisely identifying medicinal plants contribute to identifying bioactive compounds that can lead to the development of new, more effective pharmaceutical drugs.
- **Support for scientific research:** A reliable, automated system for identifying plants helps botany, scientists and pharmacologists in a study and classification of plant species, reduce manual efforts and human mistakes.
- **Maintaining biodiversity:** Automated identification tools can help monitor and document rare or endangered plant species, promoting efforts to protect the environment and biological diversity.
- **Powering of rural and distant communities:** providing available tools for plant identification

allows communities with limited access to experts or laboratories to safely and effectively benefit from traditional medicine.

1.5 Scope of the work

1. Development of a deep learning-based system for the identification of medicinal plants using leaf images.
2. Implementation of Convolutional Neural Networks (CNN), specifically using VGG19 architecture, for accurate plant classification.
3. Comparison and experimentation with other models like VGG16, MobileNet, and DenseNet to enhance model performance.
4. Improvement of prediction accuracy over traditional and existing plant identification methods
5. Exploration of real-time identification capabilities for future mobile or web-based deployment.

Chapter 2

Literature Survey

2.1 Existing Methods

Traditional methods for medicinal plant identification primarily rely on manual inspection and botanical expertise. These approaches involve analyzing morphological characteristics such as leaf shape, color, texture, and venation patterns, which are often subjective and time-consuming. Herbarium specimens and field guides are frequently used in ethnobotanical studies to match observed plants with documented species. However, these conventional techniques are limited by the availability of expert knowledge, regional biodiversity, and the inherent variability of plant structures due to seasonal or environmental factors. Moreover, manual identification becomes impractical in large-scale biodiversity surveys or in regions lacking adequate resources.

To overcome these limitations, researchers have increasingly turned toward automated techniques utilizing image processing and machine learning. Earlier computational methods employed basic feature extraction techniques like edge detection, color histograms, and texture analysis combined with classifiers such as Support Vector Machines (SVMs) or k-Nearest Neighbors (k-NN). These models, while an improvement over manual methods, often struggled with complex visual features and large intra-class variation. In recent years, deep learning—particularly Convolutional Neural Networks (CNNs)—has revolutionized plant identification by enabling models to learn hierarchical features directly from image data. Advanced architectures like VGGNet, ResNet, DenseNet, and MobileNet have significantly improved classification accuracy, robustness, and scalability, offering a practical solution for real-time and high-precision medicinal plant recognition.

2.2 Model Implementation

- **DenseNet Architecture Used:** Utilized DenseNet (Densely Connected Convolutional Networks) as the base architecture for CNN. DenseNet connects each layer to every other layer in a feed-forward manner, improving gradient flow and enabling the extraction of diverse and rich features.
- **Core Role of CNN in Image Classification:** Applied Convolutional Neural Networks (CNNs) for classification, leveraging convolutional, pooling, and fully connected layers for deep feature learning and prediction.
- **Input Image Resizing:** Conducted experiments using varying image resolutions — 64×64, 128×128, and 256×256 pixels to evaluate the model's performance across different input sizes.

- **Performance Evaluation with Accuracy Metric:** Used accuracy as the primary evaluation metric, measuring the proportion of correctly classified plant images out of the total, essential for gauging model performance.
- **Hyperparameter Tuning:** Optimized model performance through hyperparameter tuning, adjusting parameters such as learning rate, batch size, and number of epochs.
- **Medicinal Plant Dataset:** Trained the model on a curated dataset of medicinal plant leaf images, forming the foundation of the classification task.
- **Data Augmentation:** Employed data augmentation techniques like image rotation, flipping, and scaling to artificially expand the dataset and improve the model's ability to generalize.
- **Cross-Validation:** Likely implemented cross-validation to ensure the model's performance remains consistent and effective on unseen data.
- **GPU Acceleration:** Utilized GPU support to accelerate model training, especially necessary for deep learning tasks involving high-resolution images and complex models.
- **Transfer Learning via Fine-Tuning:** Fine-tuned a pre-trained DenseNet model, adapting it to the specific task of medicinal plant classification for enhanced accuracy and reduced training time.

2.3 Literature Survey

Stephen Opoku Oppong [1] This article discusses the development of OTAMNet, a Convolutional Neural Network (CNN) model, which is built upon the DenseNet architecture. The choice of DenseNet stems from its unique advantages, notably the efficient transfer of error signals to earlier layers, ensuring a robust gradient flow. Additionally, it enables the extraction of more diverse features, as each layer in DenseNet takes input from all preceding layers, in contrast to conventional CNN models where the classifier primarily relies on complex features. In a nutshell, this research offers valuable insights into the utilization of pre-trained networks for feature extraction from leaf images. It sheds light on the variations in the number of extracted features and the associated time complexity across diverse pre-trained network architectures. The decision to employ DenseNet201 as the foundation for OTAMNet is grounded in its superior performance when applied to classification tasks with various supervised learning algorithms.

Rahim Azadnia 1 [2] In this research, we explored a CNN model that exhibited remarkable precision when categorizing medicinal plant images, even when the images had different pixel dimensions. The model achieved outstanding accuracy rates, specifically 99.66% for 64x64 pixel images, 99.32% for 128x128 pixel images, and 99.45% for 256x256 pixel images. Among these dimensions, the 64x64 pixel images stood out as particularly successful, boasting an impressive overall accuracy

of 99.66%. In this case, the model made just one erroneous classification, mistaking an image from the Peppermint class for the Tulsi class. Additionally, the Bael and Stevia classes each had three images that were incorrectly categorized, while the Tulsi class had two images that were misclassified as Peppermint and Stevia. These misclassifications were attributed to the structural similarities among these plants. Interestingly, the Lemon Balm leaves achieved a flawless 100% accuracy rate, with the CNN model consistently identifying them correctly across all three-pixel dimensions. The structural resemblances among the medicinal plant leaves indeed presented a challenge, resulting in some misclassifications. Nevertheless, the proposed model demonstrated an impressive ability to make correct classifications using multiple layers of CNN architecture as shown in figure 2.1 despite the inherent similarities among the medicinal plant images. Overall, the CNN model demonstrated exceptional performance in the precise categorization of medicinal plants, especially in the 64x64 pixel dimension, with only occasional instances of misclassification. This underscores its significant potential in this particular field of study.

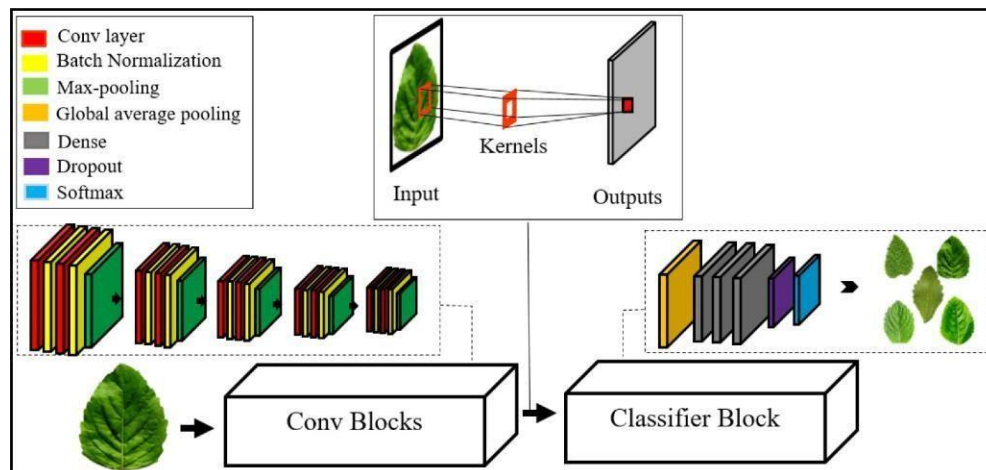


Figure 2.1: Architecture of CNN

Data Augmentation (DA) stands as a valuable strategy used to enhance the available dataset for training neural networks. Its primary aim is to elevate the model's accuracy while simultaneously reducing the risk of overfitting. In this research endeavor, we introduced five distinct DA techniques in addition to the original images. These methods encompassed four angle rotations at 45°, 90°, 135°, and 180°, along with a color manipulation approach.

Out of the entire dataset, which comprised 13,500 images, 80% (equivalent to 10,800 images) were randomly designated for training the proposed network, while the remaining 20% (totaling 2,700 images) were reserved for the purpose of testing and assessing the model's performance.

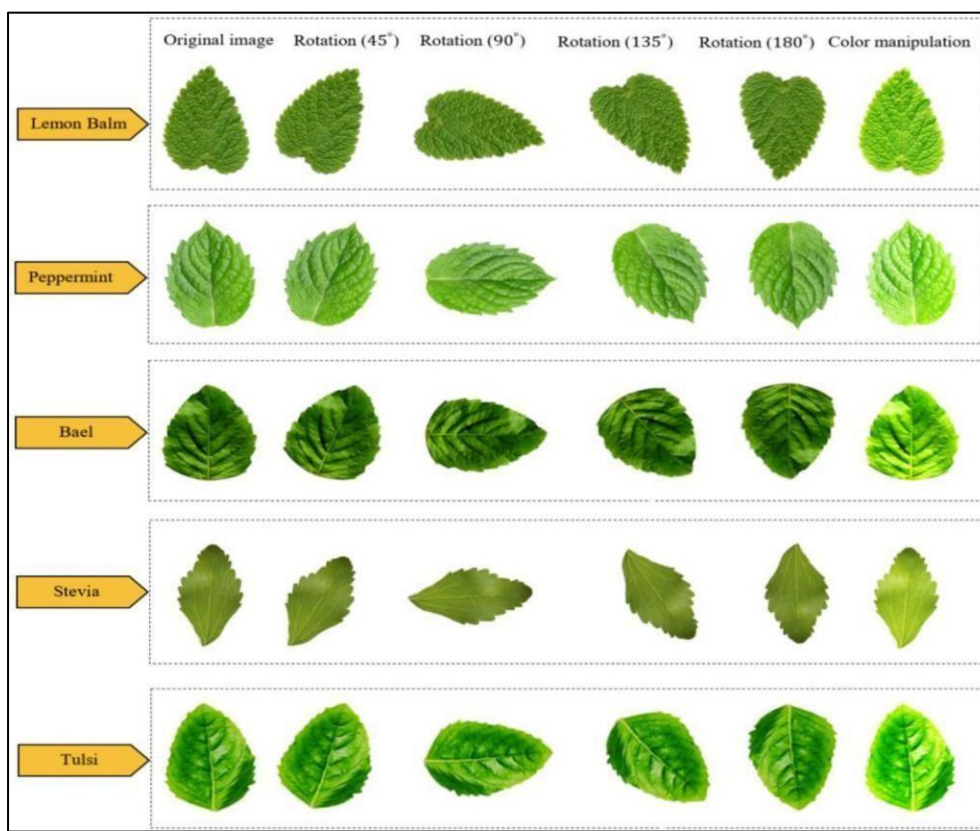


Figure 2.2: Data Augmentation Technique

2.4 Study of Algorithm from Literature Review:

Step 1: Data Collection and Preparation

Acquire a dataset of medicinal plant leaf images from the Centre for Plant Medicine Research (CPMR) in Ghana. Ensure images are in uncompressed JPEG format with YCbCr color format and dimensions of 6000x4000x3. Create a working dataset, “MyDataset,” containing 2,450 images, with 50 images from each of the 49 medicinal plant leaves.

Step 2: Data Augmentation (DA)

Apply data augmentation techniques to increase the dataset’s size. Utilize five DA methods, including 45°, 90°, 135°, and 180° angle rotations, along with color manipulation as shown in figure 2.2.

Step 3: Dataset Split

Randomly split the dataset into training and testing sets. Allocate 80% of the images (10,800) for the training set and 20% (2,700) for the testing set.

Step 4: Feature Extraction

Use ten pretrained CNN networks (AlexNet, InceptionV3, DenseNet201, GoogLeNet, ResNet101, ResNet18, ResNet50, MobileNetV2, VGG16, and VGG19) for feature extraction. Implement transfer learning to leverage features from these networks.

Step 5: Supervised Learning Algorithms

Choose six supervised learning algorithms: Naive Bayes, Support Vector Machine (SVM), K-nearest Neighbour (kNN), Decision Tree (DT), Logistic Regression (LR), and Random Forest (RF) for classification.

Step 6: OTAMNet Model Creation

Develop the OTAMNet CNN model, based on the DenseNet architecture. Incorporate handcrafted Log-Gabor features into each dense block of the DenseNet model to enhance feature extraction.

Step 7: Training and Optimization

Train the model using the training dataset with a 70:30 split. Utilize the Adam optimizer for training. Employ Google Colab servers to accelerate the training process. Implement EarlyStopping and ModelCheckpoint callbacks to prevent overfitting and save the best weights.

Step 8: Performance Evaluation

Evaluate the model's performance using accuracy and loss metrics. Calculate the true positive rate (TPR) and false positive rate (FPR) to assess classification performance.

Step 9: Cross-Dataset Evaluation

Test OTAMNet's performance on four benchmark datasets: Flavia, Swedish Leaf Dataset, MD2020, and Folio dataset. Assess its ability to generalize across diverse datasets.

Step 10: Result Analysis

Analyze and report the model's accuracy, loss, TPR, and FPR on all datasets. Highlight the model's classification performance.

Step 11: Conclusion and Findings

Summarize the findings and conclusions of the study. Emphasize the effectiveness of the proposed OTAMNet model in accurately classifying medicinal plant leaf images

Chapter 3

Software Requirement Specification

3.1 Functional Requirements

3.1.1 System Features (Functional Requirements)

The system for medicinal plant identification must fulfill the following functional requirements:

1. User Authentication:

- Users shall register and log in securely to the platform.
- The system shall validate email and password format before allowing access.
- Admins may have access to overall usage analytics.

2. Image Upload:

- Users shall upload images of plant leaves via file upload or camera input.
- The system shall validate the file format (.jpg/.png) and size (max 5MB).
- A preview of the uploaded image shall be displayed to confirm before submission.

3. Image Preprocessing & Prediction:

- The uploaded image is preprocessed (resized, normalized).
- The CNN (VGG19) model shall classify the plant based on leaf characteristics.
- The model shall return the plant species name and a confidence score.

4. Display of Information:

- Upon successful prediction, the system shall show plant details: name, uses, medicinal value, region.

- The content will be localized (English and Marathi options).

5. **Audio Output Support:**

- The platform will convert plant information into speech using Google Text-to-Speech (gTTS).
- Users may click the “Play Description” button for audible output.

6. **Multilingual Capability:**

- Users shall choose between English and Marathi languages.
- The UI and plant data content shall adapt based on selected language.

7. **History Tracking:**

- Logged-in users will see a list of past predictions with image thumbnails.
- Clicking on an entry shows full prediction details and confidence score.

8. **Logout/Session Management:**

- Users shall be logged out after a set timeout or upon request.
- The session must be securely destroyed to prevent unauthorized access.

3.2 External Interface Requirements

3.2.1 User Interfaces

- **Login Page:** Secure and simple with form validation.
- **Dashboard:** Displays upload button, recent predictions, and settings.
- **Upload Section:** Includes drag-and-drop image upload or camera capture option.
- **Prediction Result Page:** Shows plant image, name, description, and audio playback.
- **Settings:** Allows language switch, profile edit, and logout.

UI Design Considerations:

- Must be responsive across mobile and desktop.
- Use clear fonts, icons, and navigation.
- Support accessibility features (font scaling, audio).

3.2.2 Hardware Interfaces

- **Camera Access:** For real-time image capture via webcam or mobile camera.
- **Microphone/Speaker:** Required for voice-based output and possible future enhancements (e.g., voice commands).
- **Storage Access:** Image selection from local directories.

3.2.3 Software Interfaces

- **TensorFlow/Keras:** For plant image classification using the CNN model.
- **Google TTS API:** Converts text into speech for plant descriptions.
- **No-SQLAlchemy ORM:** Manages interactions with the MongoDB
- **Flask or Django Backend:** Handles model inference and UI rendering.

3.2.4 Communication Interfaces

- **RESTful API:** Manages communication between frontend and backend using JSON.
- **HTTPS Protocol:** All communication is encrypted using TLS.
- **WebSocket (optional):** For live model feedback (future scope).

3.3 Nonfunctional Requirements

3.3.1 Performance Requirements

- **System Response Time:** Predictions should complete in ≤ 5 seconds.
- **Concurrent Users:** Should support at least 100 concurrent active sessions.
- **Model Accuracy:** Must achieve $\geq 90\%$ accuracy on the validation set.
- **Availability:** 99% system uptime expected under normal conditions.

3.3.2 Safety / Security Requirements

- **Password Security:** Use salted hashes (e.g., bcrypt) for storing passwords.
- **Data Protection:** Users' uploaded images and histories must be private and encrypted.
- **Session Management:** Sessions expire after 15 minutes of inactivity.
- **Injection Protection:** SQL injection, CSRF, and XSS vulnerabilities must be mitigated.
- **Audit Logs:** Admin can track login attempts and data usage (optional future feature).

3.4 System Requirements:

3.4.1 Database Requirements

- **Database Engine:** MongoDB Atlas
- **Key Tables:**
 - Users — stores credentials, login data
 - Predictions — stores image metadata, plant result, confidence
 - Plants — stores plant name, description, and media
 - Languages — supports multilingual output

Relationships:

- One user → many predictions
- One prediction → one plant

3.4.2 Software Requirements (Platform Choice)

Client Side:

- Browser: Chrome, Firefox, Safari
- HTML5, CSS3, JavaScript
- Optional: React.js

Server Side:

- Python 3.8+
- Flask / Django framework
- TensorFlow 2.x, Keras, OpenCV
- gTTS (Google Text-to-Speech)

Database:

- MongoDB
- No-SQLAlchemy for ORM

Others:

- Deployment: AWS / Localhost
- Development Tools: VS Code, Git, Colab, Jupyter

3.4.3 Hardware Requirements**Minimum Configuration (for development or demo):**

- CPU: Intel Core i3 / AMD Ryzen 3
- RAM: 4 GB
- Disk: 500 MB free
- GPU: Not mandatory (for small datasets)
- Camera: 480p webcam or higher

Recommended Configuration (for production):

- CPU: Intel Core i5 / Ryzen 5
- RAM: 8 GB
- SSD Storage: 1 GB+
- GPU: NVIDIA GPU with CUDA support (for model retraining)
- HD camera and stereo microphone

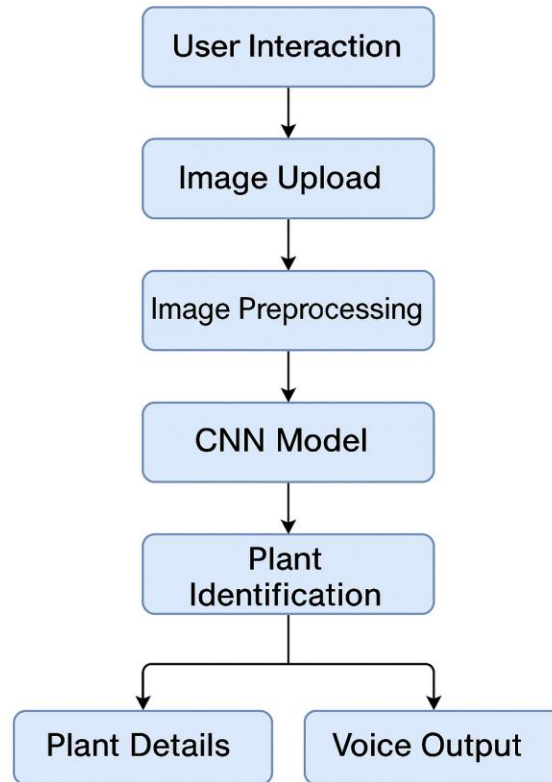


Figure 3.1: Requirement Flow diagram

The flow diagram shown in Figure 3.1 represents the process of a plant identification system using a CNN model. The system begins with User Interaction, where a user initiates the process. This is followed by the Image Upload step, where the user provides a plant image. The uploaded image then undergoes Image Preprocessing to enhance its quality and suitability for model input. Next, the CNN Model is applied to perform feature extraction and classification. Based on the model's output, the system carries out Plant Identification. Finally, the result is presented in two forms: Plant Details, which provide textual information about the identified plant, and Voice Output, which gives an audible summary for accessibility and user convenience.

Chapter 4

Proposed System

4.1 Proposed System Architecture

The proposed system architecture for medicinal plant identification is centered on the integration of Convolutional Neural Networks (CNNs) and modern image processing techniques to automate and improve the reliability of plant recognition. This system addresses the limitations of traditional plant identification methods, which are often dependent on botanical expertise and manual analysis, making them time-consuming and less accessible in remote or underserved regions.

The architecture leverages CNNs to perform automatic feature extraction and classification of medicinal plant leaves based on their shape, texture, color, and venation. The model is constructed using five convolutional blocks, each consisting of two convolutional layers with 3×3 kernels, followed by ReLU (Rectified Linear Unit) activation functions, Batch Normalization, and Max Pooling layers. This structure ensures efficient feature detection while reducing the spatial dimensions of feature maps, thus optimizing computational performance. To prevent overfitting, a Dropout layer with a rate of 0.1 is incorporated after the convolutional layers. The final classification is carried out through a fully connected classifier block, which processes the extracted features and outputs the predicted plant class.

An additional feature of the system is the integration of Google Text-to-Speech (TTS), which provides audible descriptions of the identified plant. This not only enhances user engagement but also improves the system's accessibility, particularly for visually impaired users or those preferring auditory information. By combining deep learning with assistive technologies, the system effectively bridges the gap between traditional botanical knowledge and modern computational methods, offering a scalable and user-friendly solution for medicinal plant identification across various domains, including healthcare, pharmacognosy, and environmental conservation.

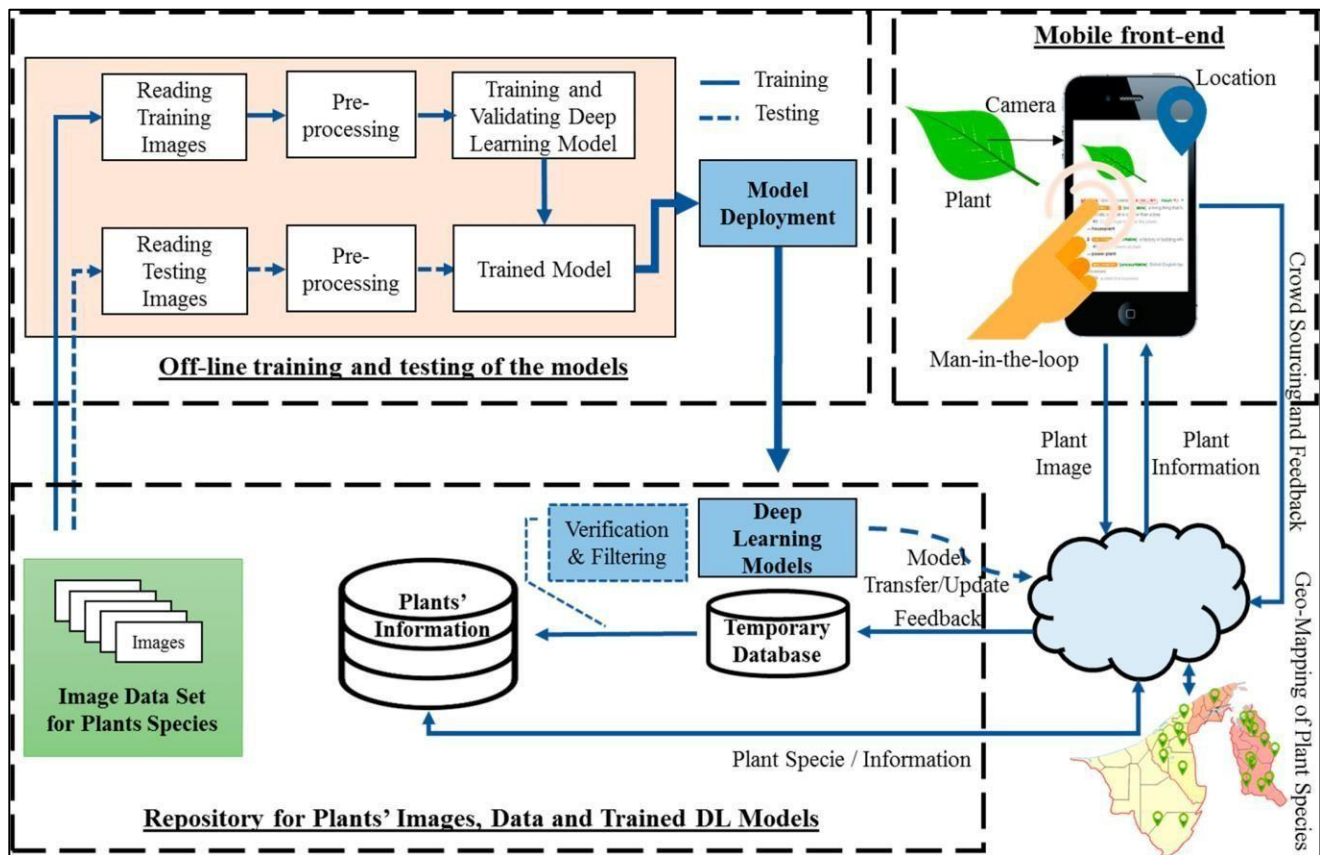


Figure 4.1: System Architecture

- Our team has developed an advanced machine vision-based solution capable of automating the real-time identification of medicinal plant species using image processing and deep learning techniques.
- We have implemented a robust Convolutional Neural Network (CNN) model based on pre-trained architectures (such as DenseNet, VGG19, etc.) to enhance accuracy in classifying medicinal plant leaves based on key features like shape, color, and venation.
- The system is designed for real-time operation, enabling plant recognition through images captured via a mobile camera or uploaded from local storage, making it highly accessible for field use.
- Our solution includes a feedback mechanism and a knowledge repository, supporting continuous learning and model improvement as more data is collected over time.

- The model supports various image input sizes (64x64, 128x128, 256x256) and demonstrates high classification accuracy across resolutions, with results validated through testing and cross-validation techniques.
- To improve generalization, data augmentation methods such as rotation, flipping, and scaling have been used during training to simulate real-world scenarios and increase dataset diversity.
- Google Text-to-Speech functionality is integrated to provide audible plant descriptions, improving accessibility for users with visual impairments or those preferring auditory interaction.
- The model has been fine-tuned using transfer learning from a pre-trained DenseNet backbone to reduce training time and improve accuracy on the medicinal plant dataset.
- The entire system is designed with scalability in mind, allowing easy integration with mobile or web-based applications for wider public usage.
- The outcome of this project is a reliable, accurate, and user-friendly medicinal plant recognition tool that bridges traditional knowledge with modern deep learning technologies, supporting both educational and healthcare domains.

4.2 Mathematical Model

The Log-Gabor function, as initially proposed by David Field, is a specialized mathematical tool designed to model and simulate the behavior of filters in the human visual system. These filters exhibit characteristics that align with the human visual system, particularly the symmetrical responses of cells when viewed on a logarithmic frequency scale. This similarity is also observed in the visual systems of mammals, highlighting the biological relevance of the Log-Gabor function

One of the key features of the Log-Gabor transform is its unique property of having an extended tail in its frequency response with no DC (direct current) component. This distinctive quality enables the creation of filters with exceptionally wide bandwidths, even extending infinitely. These broad bandwidth filters are particularly advantageous when it comes to encoding natural images, as they excel at representing the high-frequency components found in such images.

When Log-Gabor functions are evaluated on a logarithmic frequency scale, they exhibit Gaussian transfer functions. The construction of the 2D Log-Gabor filter occurs in the frequency domain due to the singularity of the logarithmic function at its origin. Within the Log-Gabor filter, radial and angular filters work together in polar coordinates to produce its overall response.

The response frequency of the radial filter can be mathematically described by the equation. This equation serves as a fundamental component in understanding and utilizing the Log-Gabor function for various applications in image processing as given in Log Gabor parameter table below in table 4.1 and visual system modeling.

Parameter	Value
Number of filter scales	11
Number of filter orientations	15
Minimum frequency	2
Scaling between center frequencies	1
Filter bandwidth	0.85
Angular spread of each filter	2.3

Table 4.1: Log Gabor Parameter

$$G(r, \theta) = \exp \left(-\frac{\log^2(r/f_0)}{(2\sigma_r^2)} - \frac{(\theta - \theta_0)^2}{2\sigma_\theta^2} \right) G(u, v) = \exp \left(-\frac{(\log(u/f_0))^2}{2\sigma_u^2} - \frac{(v - v_0)^2}{2\sigma_v^2} \right)$$

Convolutional Neural Networks (CNNs) are a class of deep neural networks designed for processing grid-like data, such as images and video. The mathematical model for CNNs consists of several key components and operations:

1) **Convolution Operation:** CNNs primarily employ convolutional layers to extract features from input data. The convolution operation can be represented as follows:

$$(f * g)(x, y) = \sum_{[i, j]} f(i, j) * g(x - i, y - j)$$

Here, $(f * g)(x, y)$ represents the result of convolution of two functions f and g , and $[i, j]$ signifies the summation over the appropriate range.

2) **Activation Function:** Common activation functions include the Rectified Linear Unit (ReLU), which can be represented as:

$$ReLU(x) = \max(0, x)$$

3) **Pooling Operation:** Pooling layers downsample the feature maps. A common operation is max pooling, represented as:

$$MaxPooling(x, y) = \max(f(x, y), f(x + 1, y), f(x, y + 1), f(x + 1, y + 1))$$

4) **Fully Connected Layer:** Fully connected layers connect all neurons from the previous layer to all neurons in the current layer. The output can be computed as:

$$y = (Wx + b)$$

5) **Loss Function:** The loss function computes the difference between the predicted output and the actual target, e.g., Mean Squared Error (MSE) for regression tasks or Cross-Entropy for classification tasks. It can be expressed as:

$$L(y, y') = -\sum_i [y_i * \log(y'_i)]$$

6) **Optimization Algorithm:** CNNs use optimization algorithms like Stochastic Gradient Descent (SGD), Adam, or RMSprop to update the network's parameters during training. A typical update step can be written as:

$$new_w = old_w - \eta * J'(old_w)$$

4.3 Algorithm

Step 1: Data Collection and Preparation

- Collect a diverse set of images of medicinal plants (leaves, flowers, roots, etc.).
- Organize the dataset into labeled classes based on plant species.

Step 2: Data Preprocessing

- Resize and normalize all images to the input size required by VGG-19.
- Apply data augmentation techniques such as:
 - Rotation
 - Scaling
 - Flipping
 - Croppingto increase dataset diversity and reduce overfitting.

Step 3: Model Selection and Pretrained Model Loading

- Load the **VGG-19** pretrained model using TensorFlow or PyTorch.
- Remove the original classification layer.
- Add a new classification layer matching the number of plant species.

Step 4: Model Training

- Split the dataset into **training**, **validation**, and **test** sets.
- Fine-tune the VGG-19 model using **transfer learning**.
- Use appropriate:
 - **Loss function**: Categorical Cross-Entropy
 - **Optimizers**: Adam or Stochastic Gradient Descent (SGD)
- Monitor validation accuracy and loss to avoid overfitting.
- Adjust **hyperparameters** such as learning rate, batch size, and epochs as needed.

Step 5: Model Evaluation

- Evaluate model performance on the **test set** using metrics:
 - Accuracy
 - Precision
 - Recall
 - F1 Score
- Analyze the **confusion matrix** to identify misclassified plant species.

Step 6: Post-Processing and Interpretation

- Map predicted class probabilities to plant species names.
- Display prediction results along with confidence scores.

Step 7: Deployment

- Deploy the trained model into a real-time system (mobile/web app).
- Allow users to upload or capture plant images for live identification.
- Integrate **Google Text-to-Speech** to provide voice output for plant names and descriptions.

Step 8: Continuous Improvement

- Continuously update the model with new images to enhance prediction accuracy.
- Experiment with improved architectures (like DenseNet, MobileNet).
- Stay updated with recent developments in CNN and transfer learning for better performance.

4.4 Complexity of Project

The complexity of a medicinal plant identification system built on deep learning goes beyond just model architecture. It includes the computational demands of training CNNs, the algorithmic structure of the classification process, and the challenges in implementation and resource planning. A comprehensive assessment ensures the system remains scalable, efficient, and easy to maintain across diverse deployment environments.

4.4.1 Computational Complexity

Training deep convolutional neural networks (CNNs) like VGG-19 is computationally intensive due to multiple layers of convolutions, pooling, and dense layers.

- **Training Time:** Considerable, especially on large image datasets. GPU acceleration (NVIDIA RTX/Tesla) significantly reduces this time.
- **Memory Usage:** High memory requirements (~16GB RAM) due to large model parameters and batch image processing.
- **Time Complexity:** The worst-case time complexity of CNN operations, considering convolution over all input channels and filters, is $O(n^3)$. This is optimized using smaller kernel sizes, reduced depth, and layer pruning.
- **Optimization Techniques:** Early stopping, dropout layers, and batch normalization are applied to reduce computational overhead.

4.4.2 Algorithmic Complexity

Though the core identification process relies heavily on CNNs, auxiliary components like image pre-processing and feature mapping also contribute to overall complexity.

- **CNN Model:** Feature extraction using VGG-19 has a layer-wise complexity that aggregates to $O(n^3)$ depending on input resolution and filter depth.

- **Image Pre-processing:** Operations such as resizing, normalization, and augmentation are linear in time complexity — $O(n)$.
- **Prediction Step:** The forward pass during inference is relatively less complex due to optimized and trained weights.

The complexity increases linearly with input size, making the model computationally heavier on high-resolution images unless scaled down intelligently.

4.4.3 Implementation Complexity

Building a real-time plant identification system involves managing dependencies across the backend (Python, TensorFlow), frontend (React/Flutter), and APIs.

- **Lines of Code:** ~2000+ (including model training scripts, UI logic, API endpoints, and deployment configs).
- **Libraries Used:** Over 20 libraries and frameworks including TensorFlow, OpenCV, Flask, NumPy, Matplotlib, and React.
- **Architecture Design:** Moderate to high complexity — the system follows a modular architecture with components such as:
 - Image upload/capture module
 - CNN-based prediction engine
 - Feedback and learning module
 - Knowledge repository interface
- **Integration:** REST APIs are used for seamless communication between backend and frontend, ensuring flexibility in mobile/web platform deployment.

4.5 Design with UML Diagrams

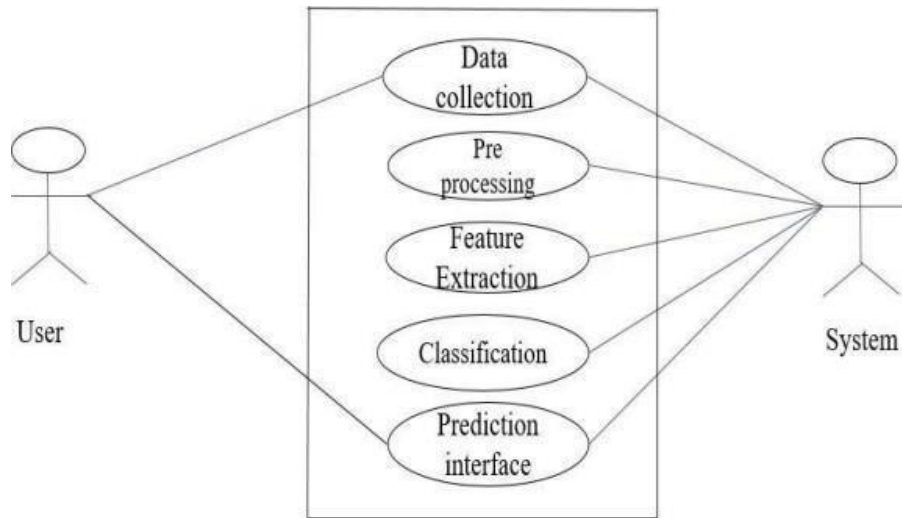


Figure 4.2: Current UML Diagram

Chapter 5

Project Plan

5.1 Project Cost Estimation

Developing a medicinal plant identification system using deep learning, especially with CNNs like VGG-19 or custom architectures, involves various hardware, software, and operational costs. These include model training resources, data preprocessing infrastructure, and deployment environments. The estimation is divided into computational and software/integration costs, with flexibility based on deployment (local/cloud-based)

5.1.1 Computational Costs

For efficient model training and real-time plant identification, the system must be capable of handling GPU-accelerated training, memory-intensive preprocessing, and image-based inference. Real-time prediction requires adequate bandwidth and responsive hardware during deployment.

Table 5.1: Computational Cost Estimate

Component	Specification	Estimated Monthly Cost (INR)
Processing Power	GPU-enabled system (e.g., NVIDIA RTX 3060/3080)	₹8,000 – ₹12,000
RAM/Memory	16 GB DDR4 RAM or above	₹1,800 – ₹3,000
Storage	512 GB SSD for fast image loading/training	₹1,200 – ₹1,800
Internet/Network	High-speed internet for dataset/API usage	₹1,000 – ₹2,000

Subtotal (Computational): ₹12,000 – ₹18,800

5.1.2 Software and Integration Costs

These involve tools and cloud services used for deploying models, text-to-speech integration, and optimization. Since most frameworks used are open source (TensorFlow/Keras), licensing costs are zero. However, cloud deployment, storage access, and API calls may introduce variable costs.

Table 5.2: Software & Integration Costs

Component	Detail	Estimated Monthly Cost (INR)
TTS Integration	Google Text-to-Speech API usage	₹500 – ₹1,000
Cloud Deployment	Hosting model/app via AWS/GCP/Heroku	₹1,000 – ₹2,000
Model Optimization	Transfer learning & pruning (manual optimization)	₹0 (effort-based cost only)
Database/Storage	Firebase/MongoDB or cloud storage for plant info	₹1,000 – ₹1,500

Subtotal (Software/Integration): ₹2,500 – ₹4,500

5.1.3 Final Cost Estimate

The final project cost estimate depends on usage frequency and deployment choice. Local deployment can reduce expenses significantly, whereas cloud-based solutions offer better scalability but at a higher recurring cost.

Table 5.3: Final Monthly Cost Estimate

Category	Minimum (INR)	Maximum (INR)
Computational Costs	₹12,000	₹18,800
Software/Integration	₹2,500	₹4,500
Total	₹14,500	₹23,300

5.2 Risk Management

Risk management is an essential component of developing and deploying a robust medicinal plant identification system based on CNNs. The integration of deep learning techniques, real-time image inputs, and cloud deployment introduces various technical, operational, and financial risks. A structured risk management framework ensures potential challenges are proactively identified, analyzed for severity, and mitigated to ensure reliability, performance, and long-term sustainability.

The risk management framework involves three main components:

- **Risk Identification:** Listing all potential risks across development and deployment stages.
- **Risk Analysis:** Assessing the likelihood and impact of each risk.
- **Risk Mitigation, Monitoring, and Management:** Implementing preventive and corrective measures.

5.2.1 Risk Identification

Key risks identified during the lifecycle of the plant identification system include:

- **Dataset Limitations:**
 - Limited or imbalanced image data could reduce model generalizability.
 - Poor-quality or mislabeled leaf images may impact model training accuracy.
- **Model Overfitting/Underperformance:**
 - CNNs might perform well on the training dataset but poorly on real-world test images.
 - Variations in lighting, orientation, and image quality could affect prediction accuracy.
- **Hardware and Infrastructure Failures:**
 - GPU/server downtime may interrupt training or hinder large-batch processing.
 - Local system crashes without proper checkpoints could result in data/model loss.
- **Security and Privacy Risks:**
 - If the system is deployed online, unauthorized API access or image data leakage can occur.
 - Use of open-source libraries without proper validation may introduce vulnerabilities.
- **Cost Overrun Risks:**
 - Cloud GPU usage, model storage, or high API requests may exceed budget if not monitored.
- **User Interface & Adoption Issues:**
 - Non-intuitive UI or unresponsiveness in mobile/web platforms could reduce usability.

5.2.2 Risk Analysis

The following table outlines the likelihood and impact level of each identified risk:

Table 5.2.1: Risk Analysis

Risk	Probability	Impact	Description
Dataset Imbalance	Medium	High	Affects classification across less-represented plant classes.
Model Overfitting	High	High	Could severely reduce real-world accuracy.
GPU/Cloud Downtime	Medium	High	Interrupts training/inference; affects availability.
Security Breach (API misuse)	Low	High	Data theft or service abuse via public endpoints.
Cost Overrun (Cloud/Infra)	Medium	Medium	Can be managed but affects scalability or academic use.
UI/UX Issues	Medium	Medium	Impacts adoption rate, especially in non-technical users.

5.2.3 Risk Mitigation, Monitoring, and Management

Table 5.2.2. Risk Mitigation, Monitoring, and Management

Risk Category	Mitigation Strategy	Monitoring Method
Data-Related Risks	Augment dataset, apply stratified sampling, use data versioning tools (e.g., DVC).	Regular model validation with diverse input samples.
Model Overfitting	Use dropout, early stopping, and cross-validation; prune layers to avoid overfitting.	Monitor training/validation loss gap; AUC scores.
Infrastructure Risks	Enable checkpointing during training, use cloud services with SLAs and failovers.	Log monitoring, cloud dashboards, usage alerts.
Security Risks	Secure API endpoints with tokens, enforce rate limiting, and audit third-party packages.	Enable logging and alerts for unauthorized access.
Cost Risks	Use free-tier credits during development, set budget alerts on cloud platforms.	Cloud billing dashboards and usage meters.
UI/UX Issues	Conduct user testing, follow accessibility standards, make UI responsive and minimal.	User feedback collection and error tracking tools.

5.3 Project Schedule

5.3.1 Project Task Set

Our team followed a collaborative and modular structure throughout the development of the Medicinal Plant Leaf Identification System. With four members, we divided responsibilities based on individual strengths and areas of interest. This structure helped us maintain parallel progress across different modules, encouraged knowledge sharing, and ensured a smooth workflow from data collection to model deployment. Regular team meetings, version-controlled collaboration, and iterative review cycles contributed to the project's overall success.

Here's how we split the work:

1. **Atharv Nikam:** Designed and developed the web application with a focus on usability and responsiveness. He also contributed to the user interface of the web-based image uploader, which allows users to submit leaf images for identification. His work ensured the system was user-friendly and accessible across different devices.
2. **Harshal Naik:** Handled the Flask backend server, developed REST APIs, and managed model integration for the identification system. He was responsible for building the server-side logic, connecting the trained CNN model with the frontend, and ensuring smooth real-time predictions from uploaded leaf images.
3. **Piyush Hole:** Focused on the design, training, and optimization of the CNN model used for plant leaf classification. He handled dataset pre-processing, image augmentation, and fine-tuning of hyper parameters to achieve high accuracy in identifying medicinal plant species.
4. **Sufiya Sayyad:** Took care of project documentation, model evaluation, testing (unit and integration), and deployment support. She ensured that proper testing was conducted on various image inputs, handled report writing, and supported the deployment of the system on cloud infrastructure for scalable usage.

The task done by on weekly basis by each and every member simultaneously for this project implementation is shown in timeline chart at figure 5.5

5.3.1 Timeline Chart



Figure 5.3: Timeline Chart

Chapter 6

Software Testing

6.1 Types of Testing

6.1.1 Requirement Traceability Testing

Requirement Traceability Testing ensures that each functional and non-functional requirement of the medicinal plant identification system is tested and validated. A Requirement Traceability Matrix (RTM) is used to link each system requirement to its respective test case.

For example:

- FR-01 (Image Upload) → TC01 (Upload JPG image and check success)
- FR-04 (Multilingual Support) → TC08 (Check if details are displayed in Marathi)
- NFR-01 (Performance Requirement) → PT01 (Verify prediction response within 5 seconds)

This ensures that:

- No requirement is left untested.
- All future changes are traceable back to their original requirement.
- Testing coverage is complete and systematic.

6.1.2 Unit Testing

Unit Testing focuses on testing individual components of the system in isolation, before they are integrated. For this project, unit testing is carried out for:

1. Flask API Functions

- Predict API: Ensures the CNN model receives image data and returns a valid response.
- Audio API: Verifies the Text-to-Speech service returns correct audio.

2. Image Preprocessing Functions

- Leaf image resizing, normalization, and augmentation functions are tested using test images.

3. Prediction Model

- The VGG19-based CNN model is tested with known images to ensure consistent classification.

4. **Voice Module**

- Ensures that text is properly converted to audio using Google TTS.

Tools Used:

- PyTest for Flask and preprocessing logic
- Unittest/TensorFlow Test Utils for model-level functions

6.1.3 Integration Testing

Integration Testing validates that the modules work correctly together. In this system, various integrations are tested, such as:

1. **Frontend (HTML/JS) ↔ Backend (Flask)**

- Image upload from UI sends request to Flask and receives a response.

2. **Backend ↔ CNN Model**

- Flask must pass the image to the model and return the correct output to the user interface.

3. **Backend ↔ Google Text-to-Speech API**

- Verifies that plant information is correctly passed to the TTS engine and that an audio file is generated.

4. **Language Toggle ↔ Description Renderer**

- Ensures that switching the language from English to Marathi updates both text and audio output.

Test Cases Include:

- Upload integration
- Audio playback after prediction
- Language change propagation across modules

6.1.4 Regression Testing

Regression Testing ensures that new changes do not break previously working features. Every update (UI change, model retraining, or new feature addition) is followed by regression tests.

What's covered:

1. **UI Regression**

- Ensure the upload button, dashboard elements, and multilingual toggles still work after UI changes.

2. Prediction Regression

- Confirm that plant predictions remain consistent and accurate even after retraining the CNN model.

3. Audio Output Regression

- Ensure the audio output still functions after adding new plants or updating plant descriptions.

4. Login/Logout Flow

- Ensure session logic is unaffected by new UI features or security updates.

5. Database Regression

- Check if previously stored predictions and user data are retrievable after database schema updates.

6.2 Test Cases and Results

6.2.1 Unit Testing

Test Case ID	Module/Feature	Test Description	Input	Expected Output	Status
UT01	Login	Valid login	user@example.com / 123456	Redirect to dashboard	Pass
UT02	Login	Invalid login	wrong@example.com / 0000	Show 'Invalid Credentials'	Pass
UT03	Image Upload	Valid image format	Upload Tulsi.jpg	Image accepted	Pass
UT04	Image Upload	Unsupported file type	Upload tulsi.pdf	Show 'Invalid file type'	Pass
UT05	CNN Prediction	Check prediction logic	Valid leaf image	Predicted class label shown	Pass
UT06	Audio Module	Validate audio playback	Click 'Play' on description	Audio plays successfully	Pass
UT07	Logout	Validate logout functionality	Click on logout	Redirect to login page	Pass

6.2.2 Integration Testing

Test Case ID	Module/Feature	Test Description	Input	Expected Output	Status
IT01	Login + Dashboard	Check login flow	Login	Dashboard loads	Pass
IT02	Upload + Prediction	Upload image and get result	Upload image	Plant name appears	Pass
IT03	Prediction + Info Display	Check prediction info	Upload → Predict	Correct plant details	Pass
IT04	Info + Audio	Verify audio for plant	Click play	Audio plays plant info	Pass

6.2.3 Acceptance Testing

Test Case ID	Module/Feature	Test Description	Input	Expected Output	Status
AT01	Image to Plant Mapping	Upload leaf and check result	Upload leaf image	Correct plant name	Pass
AT02	Audio Feature	Click 'Play'	Click on Play	Audio plays	Pass
AT03	Multilingual Support	Switch to Marathi	Select Marathi	Info in Marathi	Pass
AT04	Usability	Navigate the app	Navigate app	Intuitive UI	Pass

6.2.4 Requirement Testing

Test Case ID	Module/Feature	Test Description	Input	Expected Output	Status
RT01	FR-01	Image must be JPG/PNG	Upload JPG	Image accepted	Pass
RT02	FR-02	Show prediction after upload	Upload leaf	Plant name shown	Pass
RT03	NFR-01	Response time < 5s	Upload and predict	≤ 5s	Pass
RT04	FR-03	Audio playback	Click Play	Audio heard	Pass

RT05	NFR-02	Regional language support	Change to Marathi	Translation seen	Pass
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6.2.5 Performance Testing

Test Case ID	Module/Feature	Test Description	Input	Expected Output	Status
PT01	Upload + Predict	Response time	Upload image	< 5 seconds	Pass
PT02	Bulk Predictions	Load handling	100 uploads	No crash	Pass
PT03	High-res image	Model handling	Upload large image	Processes correctly	Pass

6.2.6 Regression Testing

Test Case ID	Module/Feature	Test Description	Input	Expected Output	Status
RG01	Login	Blank login check	Submit empty fields	Validation message	Pass
RG02	CNN	Blurred image handling	Upload blurred image	Low confidence warning	Pass
RG03	Audio	Marathi audio	Play in Marathi	Marathi audio heard	Pass
RG04	Upload	Large file crash	Upload 10MB image	Size limit warning	Pass

6.2.7 System Testing

Test Case ID	Module/Feature	Test Description	Input	Expected Output	Status
ST01	End-to-End Flow	Complete plant detection flow	Login → Upload → Predict	Plant details + Audio + Dashboard	Pass
ST02	File Upload & Validation	Check image type and size	Upload JPG < 5MB	Image accepted	Pass
ST03	Prediction & Accuracy	Validate model prediction	Leaf image of Tulsi	Correct plant name shown	Pass
ST04	Multilingual Flow	Check regional language output	Select Marathi → Predict	Info in Marathi displayed	Pass

ST05	Voice Output	Check audio after plant detection	Click 'Play'	Accurate plant info read aloud	Pass
ST06	User Logout	Check logout from dashboard	Click Logout	Redirect to login page	Pass

Chapter 7

Results & Discussion

7.1 Outcomes

- Successfully developed a deep learning-based model using Convolutional Neural Networks (CNN) for the classification and identification of medicinal plants through leaf images.
- Implemented and tested advanced architectures like VGG19, with comparative analysis against other models such as VGG16, MobileNet, and DenseNet.
- Achieved high accuracy in plant species prediction, demonstrating the effectiveness of CNN in extracting and learning meaningful features from leaf images.
- Improved existing identification methodologies by reducing manual intervention and increasing the speed and reliability of classification.
- Created a foundation for real-time plant identification systems, with potential applications in mobile or web platforms.
- Designed a user-friendly system concept with future scope for voice input and regional language integration to make it accessible to a broader audience, including rural and non-technical users.
- Contributed to bridging the gap between traditional plant knowledge and modern technology, supporting applications in healthcare, botany, and environmental conservation.

7.2 Results analysis and Validations

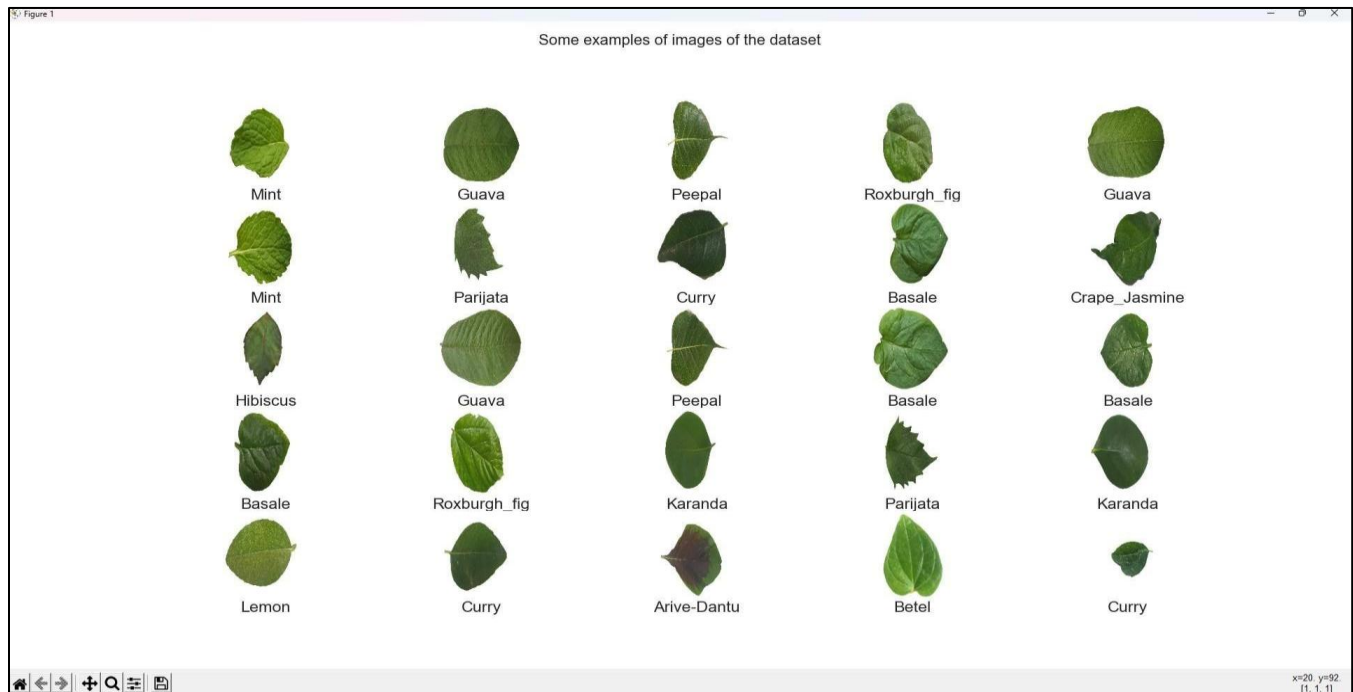


Figure 7.1: Sample Images of Medicinal Plants from dataset

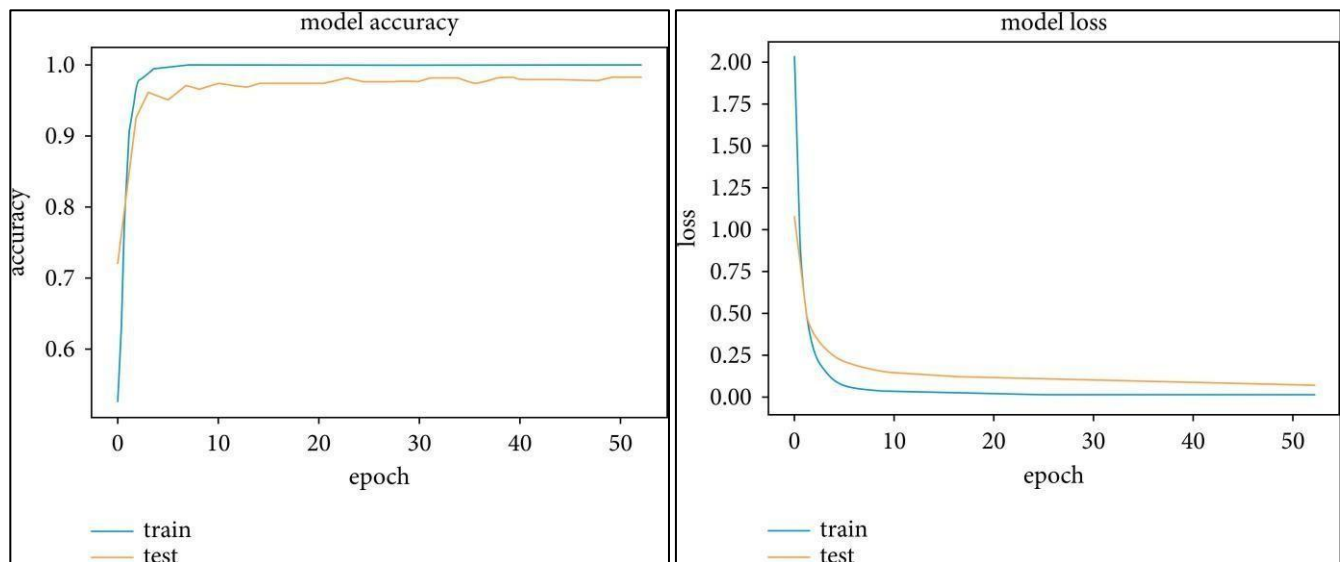


Figure 7.2: Accuracy and Loss of Trained CNN Model

Finetuned Deep Convolutional Neural Network	Test Accuracy
VGG16	90.07
VGG19	92.75
MobileNet	78.18
Xception	71.06
Alex-Net	90
Inception-V3	85

Table 7.1: Result comparison of fine-tuned DCNN models for Medicinal Plant Detection

Table 8.1 shows the test accuracy of various fine-tuned Deep Convolutional Neural Network models used for plant identification. Among them, VGG19 achieved the highest accuracy of 92.75%, followed by VGG16 with 90.07% and Alex-Net with 90%. Inception-V3 recorded 85%, while MobileNet and Xception showed relatively lower accuracies of 78.18% and 71.06% respectively, indicating that deeper architectures like VGG19 perform better for this task.

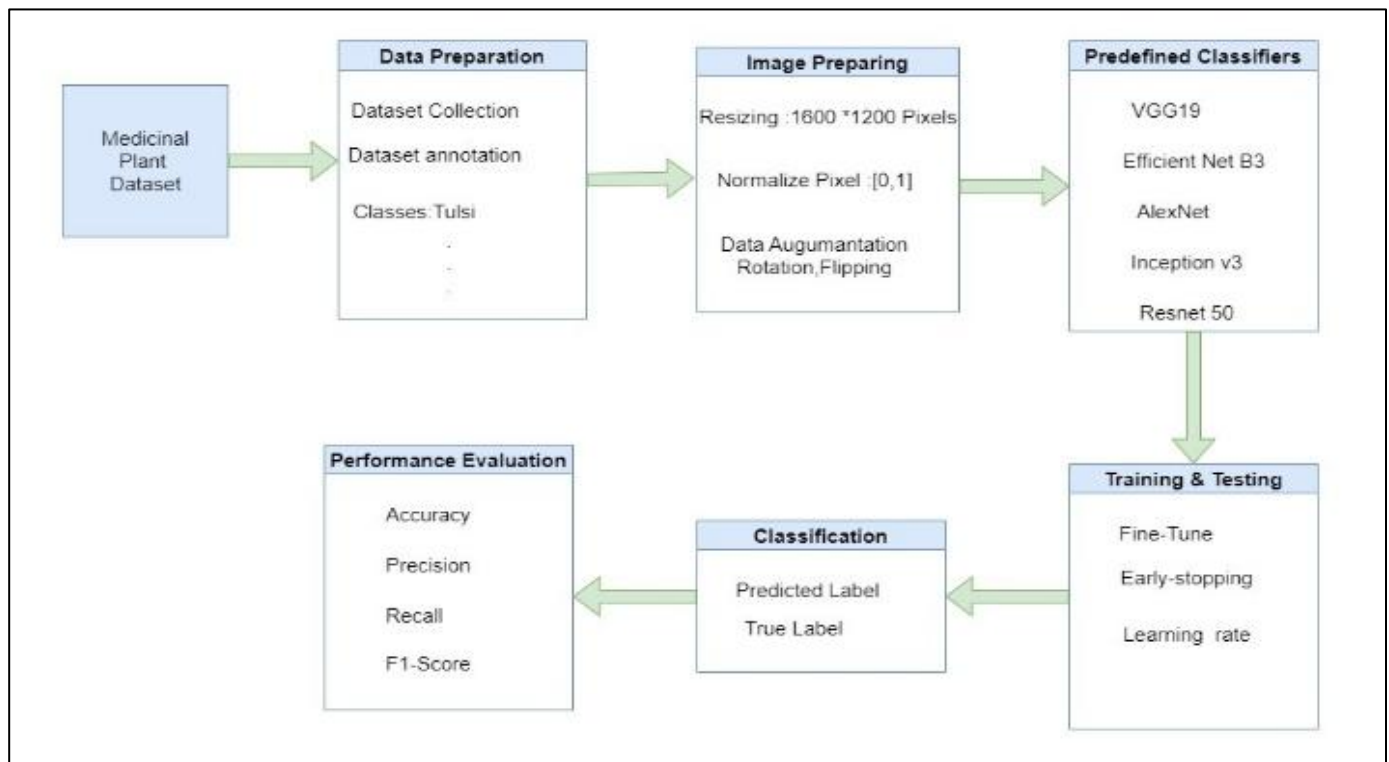
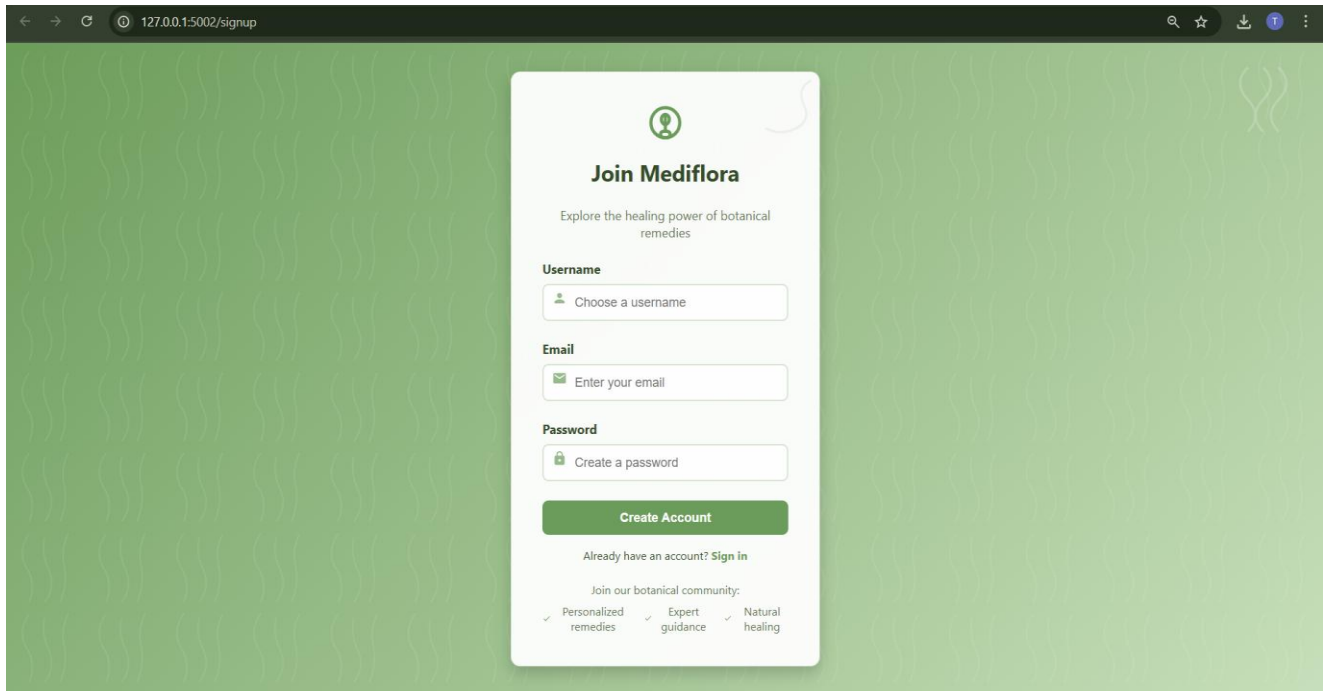


Figure 7.3: Implementation

7.3 Screenshots

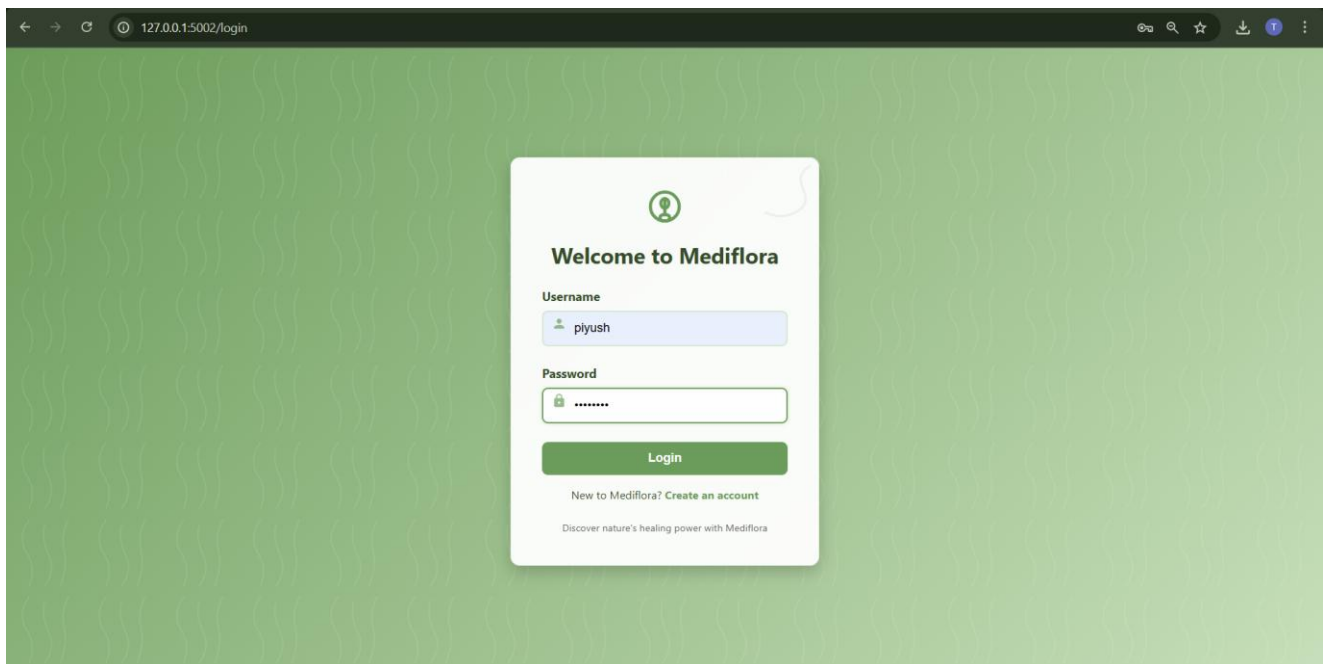
Sign Up Page :



The screenshot shows the 'Join Mediflora' sign-up page. The page has a green background with a subtle leaf pattern. A white card in the center contains the following elements:

- A green circular icon with a white leaf inside.
- Join Mediflora**
- Explore the healing power of botanical remedies
- Username**: A text input field with a placeholder 'Choose a username'.
- Email**: A text input field with a placeholder 'Enter your email'.
- Password**: A text input field with a placeholder 'Create a password'.
- Create Account**: A green button.
- Already have an account? [Sign in](#)
- Join our botanical community:
- Three checkmarks with labels: 'Personalized remedies', 'Expert guidance', and 'Natural healing'.

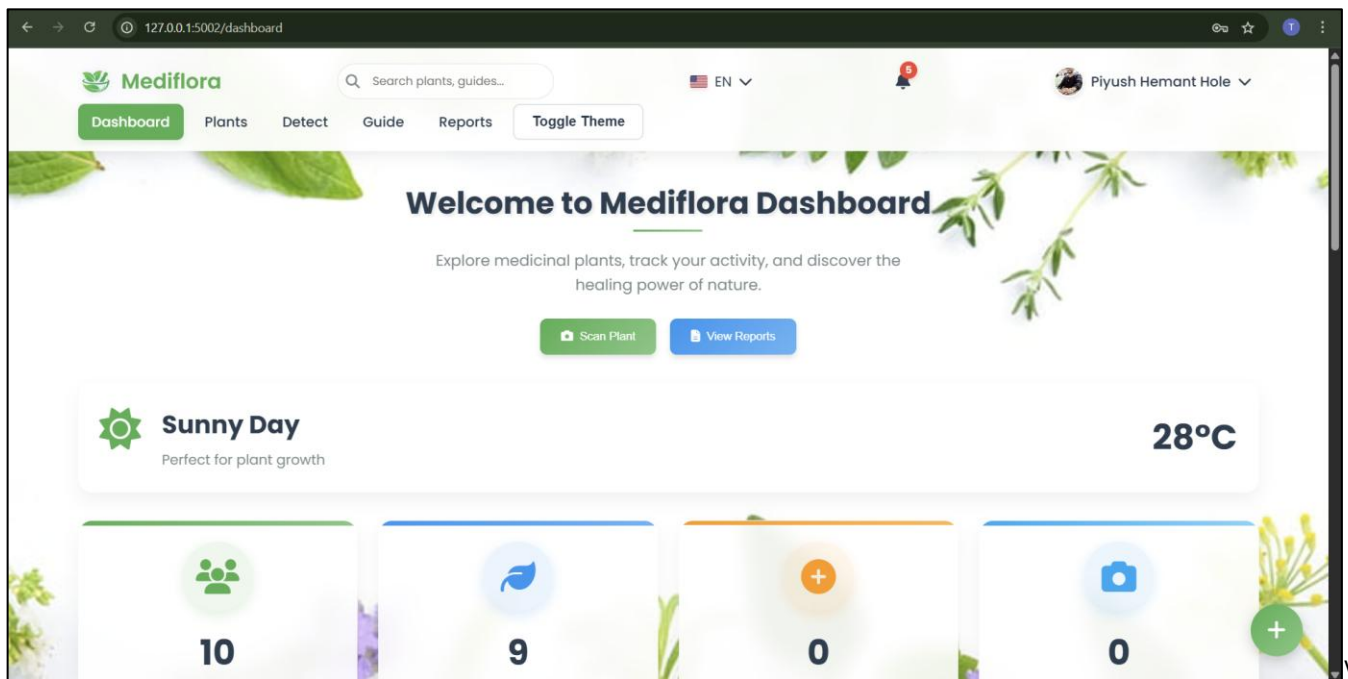
Login Page :



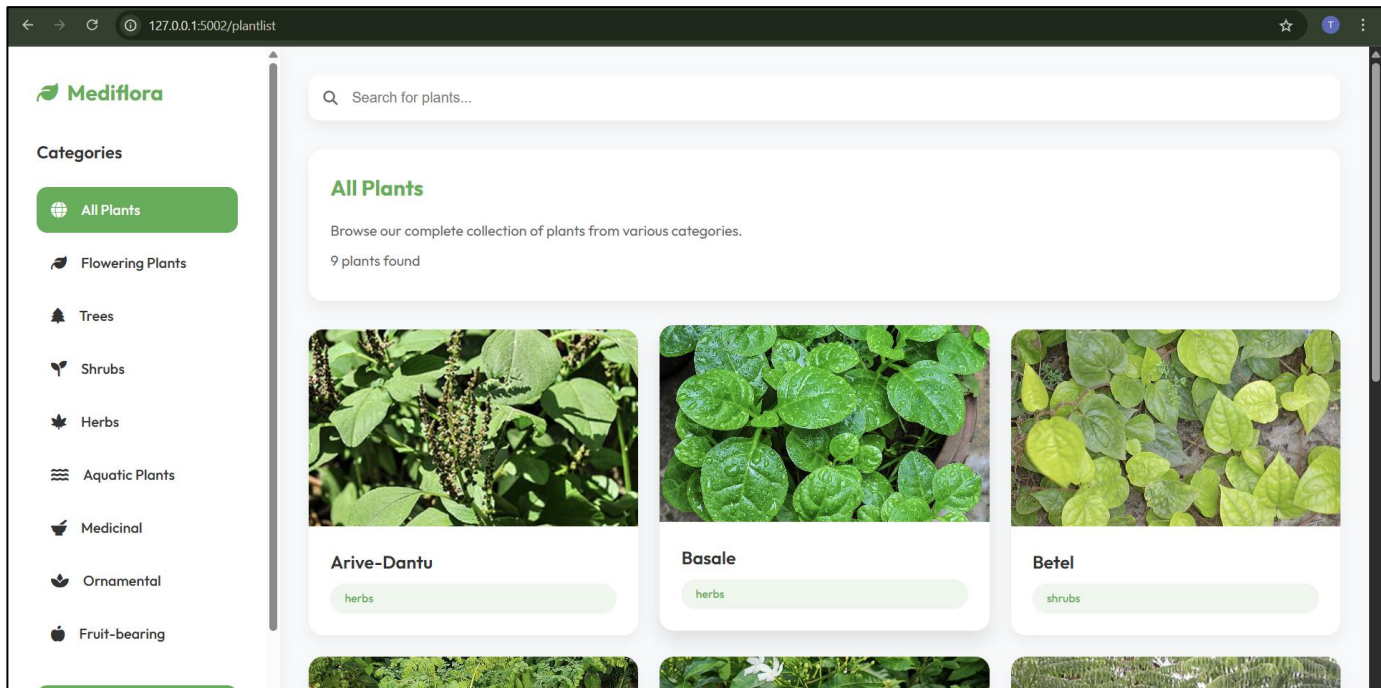
The screenshot shows the 'Welcome to Mediflora' login page. The page has a green background with a subtle leaf pattern. A white card in the center contains the following elements:

- A green circular icon with a white leaf inside.
- Welcome to Mediflora**
- Username**: A text input field with the value 'plyush'.
- Password**: A text input field with masked characters '*****'.
- Login**: A green button.
- New to Mediflora? [Create an account](#)
- Discover nature's healing power with Mediflora

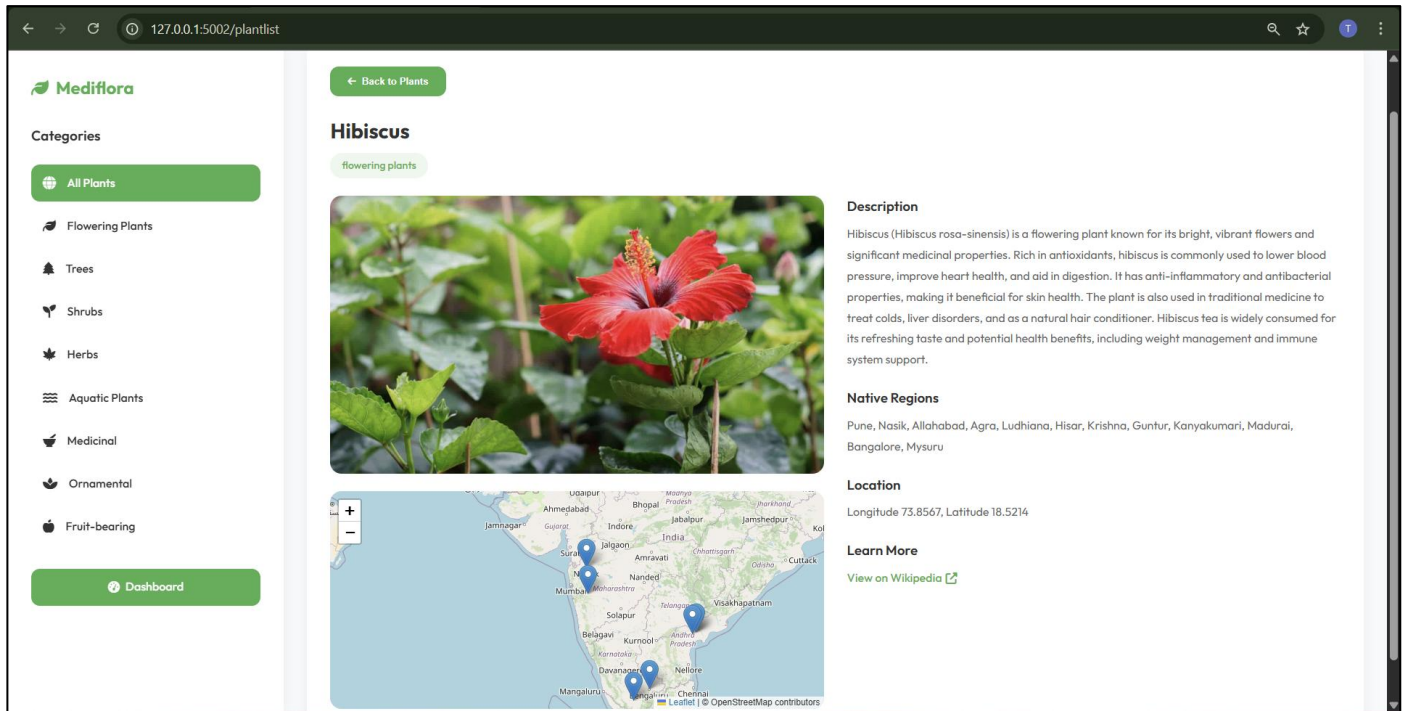
Dashboard Page :



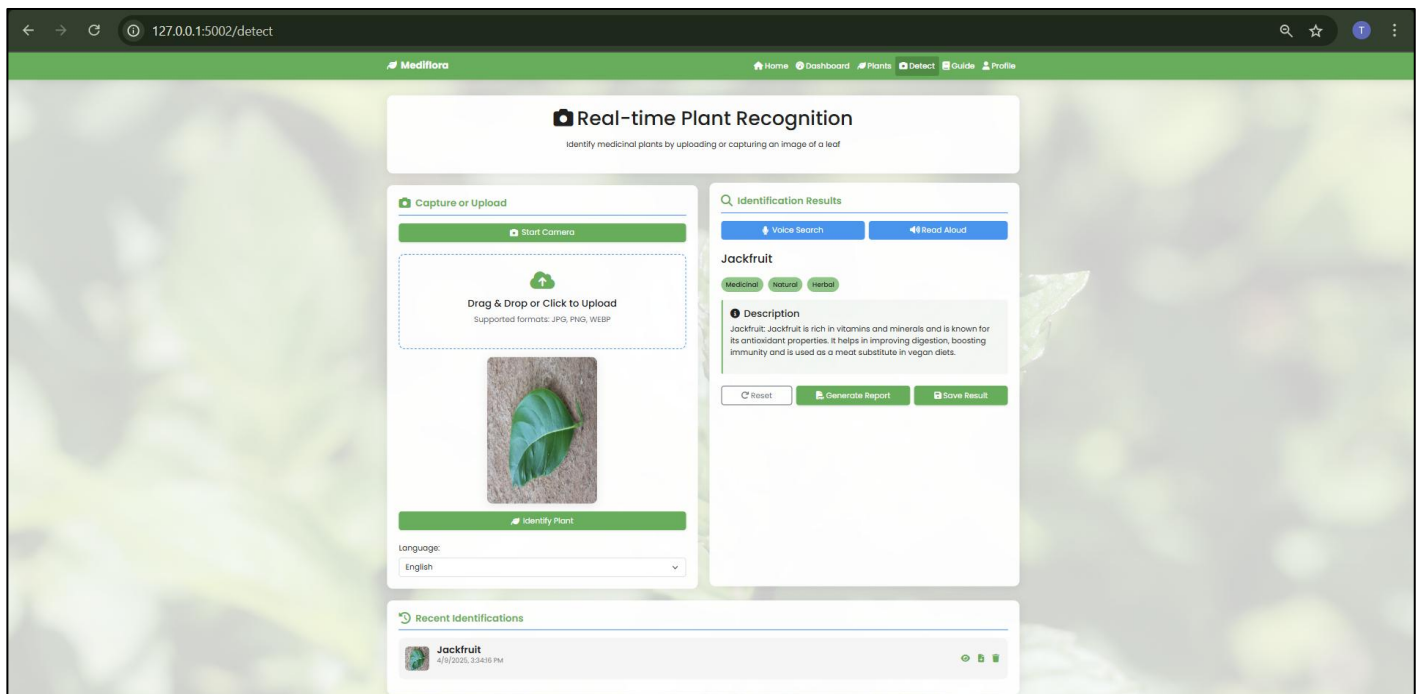
PlantList :



Plant Details :



Medicinal Plant Detection :



Multi Language Support :

127.0.0.1:5002/detect



Real-time Plant Recognition

Identify medicinal plants by uploading or capturing an image of a leaf

Capture or Upload

Start Camera



Drag & Drop or Click to Upload

Supported formats: JPG, PNG, WEBP



Identify Plant

Language:

Hindi

Identification Results

Voice Search
Read Aloud

Jackfruit

Medicinal
Natural
Herbal

Description

जैकाफ़ूट, जैकाफ़ूट विटामिन्स और मिनरल्स से भरपूर होता है और इसके एंटीऑक्सीडेंट गुणों के लिए जाना जाता है। यह पाचन में सुधार करने, प्रतिरक्षा को बढ़ाने में मदद करता है और शाकाहारी आहार में मांस के विकल्प के रूप में उपयोग किया जाता है।

Reset
Generate Report
Save Result

Details Report of Predicted Plant :

Medicinal Plant Report

Jackfruit (*Artocarpus heterophyllus*)

Plant Imagery

Plant Images


Plant image of Jackfruit

Leaves Images



Leaves image of Jackfruit

Flowers Images

Failed to load flowers image: 406 Client Error: Not Acceptable for url: <https://www.growables.org/information/TropicalFruit/Images/JackfruitWiki16.jpg>

Fruits Images


Fruits image of Jackfruit

Botanical Information

Scientific Name:	Artocarpus heterophyllus
Family:	Moraceae
Kingdom:	Plantae
Conservation Status:	Least Concern
Plant Category:	Tree

Medicinal Properties

This plant has been traditionally used for the following medicinal purposes:

- Antioxidant
- Anti-inflammatory
- Antimicrobial
- Digestive aid
- Anti-cancer

Traditional Uses

Historical and cultural applications include:

Chapter 8

Contribution to Sustainable Development Goals

8.1 Introduction to SDGs

The Sustainable Development Goals (SDGs), established by the United Nations in 2015, are a universal call to action to end poverty, protect the planet, and ensure peace and prosperity for all by the year 2030. There are 17 interlinked goals and 169 associated targets, each designed to address specific global challenges — from climate change and economic inequality to education, health, and environmental sustainability.

The SDGs are not isolated ambitions but are deeply interconnected; progress in one goal often facilitates progress in others. They serve as a blueprint for governments, organizations, and individuals to drive meaningful change. Each goal is supported by measurable indicators and aims to promote inclusive, equitable, and sustainable development for current and future generations.

Technological innovation, scientific research, and digital transformation are considered essential enablers in achieving many of these goals. In this context, AI-driven solutions — especially in fields like agriculture, biodiversity, and environmental conservation — are playing a pivotal role in promoting sustainable ecosystems and preserving natural resources.

8.2 Mapping of the Project to Relevant SDGs

SDG 15 – Life on Land is dedicated to protecting, restoring, and promoting the sustainable use of terrestrial ecosystems, managing forests responsibly, combating desertification, halting land degradation, and preventing biodiversity loss. This goal is critical in ensuring the balance and sustainability of natural ecosystems that support life on Earth.

Relevance to Project:

The project is directly related to SDG 15, which aims to protect, restore, and promote sustainable use of terrestrial ecosystems and halt biodiversity loss. Medicinal plants are a critical part of biodiversity and traditional medicine systems. Accurate identification is essential to ensure their preservation and sustainable use.

Impact and Contribution:

- Promotes the conservation of native and endangered plant species through AI-powered identification.
- Assists forest officials, researchers, and local communities in preventing the misidentification and overharvesting of vulnerable species.
- Supports ethical and sustainable harvesting by empowering tribal and rural communities with digital tools to recognize and use medicinal plants responsibly.
- Enables preservation of traditional knowledge by documenting medicinal plants in a digital and accessible format.
- Aids in combating illegal trade and exploitation of protected medicinal plants by providing technological support for real-time verification.

Indicators and Outcomes:

- Reduced risk of biodiversity loss by improving awareness and identification of rare species.
- Greater involvement of local communities in conservation efforts through accessible technology.
- Potential to scale to national biodiversity databases or mobile applications for real-time plant tracking.
- Promotes ecological monitoring by potentially integrating geolocation-based mapping of medicinal plant habitats in future iterations.

8.3 Project Impact Assessment

The impact of this project aligns closely with the goals and targets of SDG 15. It enables real-time identification of medicinal plants in forests and rural areas, which aids in preventing biodiversity loss through unintentional harvesting of rare or endangered plants. By promoting the use of technology in ecological monitoring, it helps integrate biodiversity values into conservation practices and community awareness programs.

Key impacts include:

- **Environmental Impact:** Helps conserve rare and native plant species by accurately identifying them and supporting their protection.
- **Social Impact:** Empowers local communities, including tribal and rural populations, by providing them with easy-to-use tools for identifying and sustainably using medicinal plants.
- **Cultural and Knowledge Preservation:** Supports the recognition and preservation of traditional medicinal knowledge systems by documenting and verifying plant species scientifically.
- **Sustainable Practices:** Reduces ecological pressure on land by preventing overharvesting and encouraging the sustainable use of natural resources.

Through these outcomes, the project fosters responsible land use, supports ecosystem balance, and promotes the ethical and sustainable use of terrestrial biodiversity — directly contributing to the achievement of SDG 15.

Chapter 9

Conclusion

9.1 Conclusion

The project titled “Identification of Medicinal Plants Using Image Processing with Voice Technology” stands as a significant advancement in the interdisciplinary domain of plant sciences, artificial intelligence, and assistive technology. By integrating deep learning models (VGG-19) for accurate image-based classification and incorporating voice technology for interactive user engagement, the system bridges the gap between traditional medicinal knowledge and modern digital tools.

Key Achievements and Outcomes:

1. Precise Plant Identification:

- The system successfully automates the identification of medicinal plants based on leaf images, utilizing advanced feature extraction capabilities of the VGG-19 model.
- It reduces human error, enhances consistency, and significantly lowers the dependency on expert knowledge for plant recognition.

2. Voice-Assisted Technology:

- With built-in speech recognition and audio output functionalities, the application promotes ease of use and inclusivity.
- Users can receive medicinal plant details and usage information through voice output, which improves accessibility for visually impaired users or individuals with low literacy levels.

3. Educational and Research Utility:

- The tool serves as an effective educational aid for students in botany, Ayurveda, pharmacy, and related domains.
- It has potential applications in rural healthcare outreach programs, traditional knowledge preservation, and environmental awareness campaigns.

4. Technology Integration:

- The use of **convolutional neural networks (CNNs)**, a **Flask-based backend**, and a responsive **user interface** ensures smooth user interaction and quick feedback.
- The modular and scalable design allows for integration with mobile apps, web platforms, and future IoT-based plant monitoring systems.

5. Support for Biodiversity and Sustainability:

- By cataloging and educating users about local and endangered medicinal plants, the system encourages responsible use and conservation efforts.
- It contributes to the preservation of indigenous knowledge systems, aligning with goals of sustainable development and environmental stewardship.

Broader Implications:

This project not only addresses a technological challenge but also contributes meaningfully to the preservation of traditional medicine, promotion of biodiversity, and democratization of scientific tools. It opens doors for real-world deployment in tribal communities, agriculture departments, herbal research centers, and public health initiatives.

Additionally, the project provides a solid foundation for future enhancements such as:

- Real-time field identification via mobile cameras.
- Multi-language support for vernacular users.
- Integration with plant usage databases for personalized recommendations.
- Augmented reality (AR) to visualize plant effects or habitat data.

Final Thoughts:

The convergence of AI, natural language processing, and plant biology within this project reflects the power of interdisciplinary collaboration. By transforming a traditional task into a digital experience, this initiative not only enhances public access to plant-based knowledge but also encourages curiosity, conservation, and innovation. It lays the groundwork for a new generation of tools that are intelligent, inclusive, and impactful.

9.2 Future Work

While this project has successfully laid the groundwork for the intelligent identification of medicinal plants using image processing and voice technology, several enhancements and expansions are envisioned to amplify its accuracy, usability, and societal impact.

1. Improved Accuracy and Model Optimization

- **Refinement of Deep Learning Models:** Future iterations aim to fine-tune the current VGG-19 model and explore more efficient architectures such as ResNet or EfficientNet to improve performance and reduce inference time.
- **Ensemble Learning Techniques:** Combining predictions from multiple classifiers using ensemble approaches can enhance model reliability, especially for difficult-to-identify plant species.

3. Dataset Expansion and Rare Species Inclusion

- **Larger and Diverse Dataset:** Expanding the dataset with high-quality images captured under various conditions will improve model generalization and robustness.
- **Inclusion of Rare Species:** Collaborating with botanical experts and herbal research institutions to include rare or endangered medicinal plants will add significant value to the system.

3. Multilingual Voice Technology Enhancement

- **Support for Regional Languages:** Enhancing the voice recognition system to support multiple Indian and global languages will ensure broader accessibility.
- **Accent Adaptability:** Training the voice model to better handle diverse pronunciations and speech variations will improve the overall user experience.

4. Web App Optimization for Mobile

- **Cross-Device Compatibility:** Although the current system is a web application, future efforts will focus on ensuring full responsiveness and smooth performance across all mobile devices and browsers.
- **Mobile-First Design:** Improving the UI/UX for mobile users will ensure seamless plant identification on-the-go without the need for a separate native mobile application.

5. Data Security and User Feedback

- **Security and Privacy Measures:** Implementing secure data handling protocols will ensure protection of any user-uploaded images or voice queries.
- **Feedback and Learning Mechanism:** Introducing a user feedback system will allow corrections and improvements in plant identification, contributing to ongoing model refinement.

6. Educational and Community Engagement

- **Collaboration with Indigenous Communities:** Respectfully integrating indigenous knowledge on plant uses can enhance the dataset and preserve traditional medicine.
- **Awareness and Learning Tools:** Creating interactive learning resources and digital campaigns can promote the importance of medicinal plant conservation, especially among students and researchers.

7. Long-term Maintenance and Scalability

- **Cloud-Based Deployment:** Hosting the system on scalable cloud platforms (e.g., AWS, GCP) ensures high availability and performance as usage scales.
- **Periodic Model Updates:** Regular retraining of the model with newly acquired data will keep the system relevant and up-to-date.

9.3 Applications

The system involving the identification of medicinal plants using image processing and potentially voice technology has a wide range of practical applications in various fields:

- **Healthcare and Medicine:**

- Medicinal Plant Identification: Healthcare professionals can use this technology to accurately identify medicinal plants, ensuring the correct usage of herbal remedies in treatments.
- Herbal Medicine Quality Control: Verify the authenticity and quality of herbal medicines by identifying the plant species used in their production.

- **Pharmaceutical Industry:**

- Drug Discovery: Assist pharmaceutical companies in discovering new drugs or compounds from medicinal plants.
- Quality Assurance: Ensure the correct plant materials are used in the production of pharmaceutical products derived from herbs.

- **Botanical Research and Conservation:**

- Plant Species Identification: Aid botanists and researchers in identifying and documenting plant species, including rare and endangered medicinal plants.
- Conservation Efforts: Help in the conservation of endangered medicinal plant species by accurately identifying them in their natural habitat.

- **Herbal and Alternative Medicine:**

- Herbalists and Practitioners: Assist herbalists and alternative medicine practitioners in plant identification for preparing traditional remedies.
- Public Awareness: Educate the public about the benefits and usage of medicinal plants for holistic healthcare.

- **Agriculture and Horticulture:**

- Plant Breeders: Support plant breeders in developing new varieties of medicinal plants with specific properties.
- Gardeners and Horticulturists: Help gardeners and horticulturists identify and care for medicinal plants in gardens and nurseries.

- **Food and Beverage Industry:**

- Culinary Use: Aid chefs and culinary experts in identifying edible and medicinal herbs for culinary purposes.
- Beverage Production: Ensure the use of correct plant ingredients in the production of herbal teas and infusions.

- **Education and Training:**
 - Botanical Education: Serve as an educational tool for students and enthusiasts interested in botany and plant sciences.
 - Training for Healthcare Professionals: Provide training modules for healthcare professionals to enhance their knowledge of herbal medicines.
- **Consumer Apps and Mobile Platforms:**
 - Mobile Apps: Develop user-friendly mobile applications for consumers interested in identifying medicinal plants in their surroundings.
 - Citizen Science: Encourage citizen scientists to participate in plant identification and data collection
- **Environmental and Ecological Studies:**
 - Ecosystem Analysis: Contribute to the assessment of the ecological impact of medicinal plants on ecosystems and biodiversity.
 - Invasive Species Detection: Help detect and manage invasive medicinal plant species in natural environments.
- **Traditional Knowledge Preservation:**
 - Indigenous Communities: Collaborate with indigenous communities to document and preserve traditional knowledge of medicinal plants.
 - Cultural Heritage: Contribute to the protection of cultural and traditional heritage related to herbal remedies.

The system's applications are diverse, ranging from healthcare to environmental conservation and education. The technology can provide valuable support in ensuring the proper use and preservation of medicinal plants, as well as expanding our knowledge of the natural world.

APPENDIX

Appendix A: Details of Paper application

Paper	Title	Conference/Journal	Status
1	Detection of Medicinal Plants and its effects to cure Diseases using Image Processing with various Deep-Learning techniques	9th International Conference on Control Communication, Computing and Automation	Submitted

Paper 1: Proof of submission

9th International Conference on Control Communication, Computing and Automation : Submission (406) has been created. Inbox x

 **Microsoft CMT** <noreply@msr-cmt.org> 20:07 (26 minutes ago) ★ 😊 ↩ ⋮
to me ▾

Hello,

The following submission has been created.

Track Name: Computer Vision & ARVR

Paper ID: 406

Paper Title: Detection of Medicinal Plants and its effects to cure Diseases using Image Processing with various Deep-Learning techniques

Abstract:
Identification of medicinal herbs for traditional use, health care, and environmental conservation is necessary. Conventional identification methods are time-consuming and dependent on the specific expertise. Employing what technology can offer, an iteration of image processing and machine learning techniques is proposed to speed up and make plant recognition more reliable. The study aims its spotlight the classification of different portions of leaves by using the method of species identification. The paper uses CNNs (Convolutional Neural Networks) that underline the technique. Besides CNN architectures, this study evaluates several models' performances of advanced deep learning pretraining like VGG19, AlexNet, Resnet50, EfficientNetB3, and Inception-V3. Techniques, that are strong for one model, in identifying the medicinal plant are weak in the other models. This is established with the help of thorough experiments. This study's results could greatly better the resolution of the recognition and determination of plant species using the color and shape of leaves and deep learning algorithms as a basis for it. By making the whole identification process simpler and by the use of world-top technology, this research is going to deal with urgent health care and environmental applications for fast and reliable methods of identification. At the end of this comparative study, it is shown that the utilization of fully explored deep learning models is the very fact that gives a higher understanding of the proper application of the knowledge to the purposes of development of traditional medicine, healthcare, and protection of the environment.

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Secondary Subject Areas: Not Entered

Submission Files:

Final_Paper_1.pdf (551 Kb, Thu, 10 Apr 2025 14:37:02 GMT)

Appendix B: Plagiarism report of the project report

Final Black Book

by Piyush

General metrics

85,539	11,442	1193	45 min 46 sec	1 hr 28 min
characters	words	sentences	reading time	speaking time

Score



This text scores better than 90% of all texts checked by Grammarly

Writing Issues

321	65	256
Issues left	Critical	Advanced

Plagiarism



229
sources

4% of your text matches 229 sources on the web or in archives of academic publications

References

- [1]Rahim Azadnia, Mohammed Maitham Al-Amidi, Hamed Mohammadi, Mehmet Akif Cifci, Avat Daryab, “An AI Based Approach for Medicinal Plant Identification Using,” MDPI, pp. 11-16, 2 November 2022.
- [2]Stephen Opoku Oppong, Frimpong Twum, James Ben Hayfron-Acquah, Yaw Marfo Missah, “Medicinal Plant Identification using Gabor Filters and Deep,” ResearchGate, no. 10.3844/jcssp.2021.1210.1221, pp. 11-13, 2021.
- [3]Nayana G. Gavhale, Dr. A. P. Thakare, “Medicinal Plant Identification using Image,” IJRASET, vol. V, no. 8, pp. 2541-2547, 2020.
- [4]Amey Sunil Deshmukh, Pushppavisha Mani Mudhaliar, Dr. Surabhi Thorat, “Ayurvedic Plant Identification using Image Processing and,” IJSRCSEIT, pp. 212-218, 10 December 2021.
- [5]Manojkumar P., Surya C. M., Varun P. Gopi, “Identification of Ayurvedic Medicinal Plants by,” in ICRCRCN, Kerela, 2017.
- [6]Anupriya Varshney, Kunal Dongare, Muskan Shrivastava, Janhavi Naik, Dr. Sanjesh Pawale, “Ayurvedic Plant Species Detection System Using,” IJRASET, vol. XII, no. 10, pp. 1265-1270, 2022.
- [7]Liwen Gao, Xiaohua Lin, “A method for accurately segmenting images of medicinal plant leaves with,” ELSEVIER, pp. 426-445, 2018.
- [8]Sameerchand Pudaruth, , Mohamad Fawzi Mahomoodally, Noushreen Kissoon, Fadil Chady, “MedicPlant: A mobile application for the recognition of medicinal plants from the Republic of Mauritius using deep learning in real-time,” IAES International Journal of Artificial Intelligence (IJ-AI), vol. 10, no. 4, pp. 938-947, 2021.
- [9]S. ROOPASHREE, J. ANITHA, “DeepHerb: A Vision Based System for Medicinal Plants Using Xception Features,” IEEE Access, vol. 9, pp. 135927-135941, 2021.
- [10]Amin Nasiri, Amin Taheri-Garavand, Dimitrios Fanourakis, Yu-Dong Zhang, Nikolaos Nikoloudakis, “Automated Grapevine Cultivar Identification via Leaf Imaging and Deep Convolutional Neural Networks: A Proof-of-Concept Study Employing Primary Iranian Varieties,” MDPI, pp. 10-18, 8 August 2021.
- [11]Naeem, S.; Ali, A.; Chesneau, C.; Tahir, M.H.; Jamal, F.; Sherwani, R.A.K.; Ul Hassan, “The classification of medicinal plant leaves based on multispectral and texture feature using machine learning approach”. Agronomy 2021

- [12] Mukherjee, G.; Tudu, B.; Chatterjee, “A convolutional neural network-driven computer vision system toward identification of species and maturity stage of medicinal leaves: Case studies with Neem, Tulsi and Kalmegh leaves”. *Soft Computing* 2021
- [13] Bisen, D. “Deep convolutional neural network based plant species recognition through features of leaf”. *Multimed. Tools Appl.* 2021.
- [14] C. Szegedy, S. Ioffe, V. Vanhoucke, and A. Alemi, “Inception-V4, Inception-Resnet and The Impact of Residual Connections on Learning,” in *Proceedings of 1th International Conference on Artificial Intelligence*, pp. 4278-4284, 2016.
- [15] J. E. Luján-García, C. Yáñez-Márquez, Y. Villuendas-Rey, and O. Camacho-Nieto, “A Transfer Learning Method for Pneumonia Classification and Visualization,” *Applied Scienc*, vol. 10, no. 8, p. 2908, 2020
- [16] T. Nguyen Quoc, V. Truong Hoang, “Medicinal Plant identification in the wild by using CNN”, in: *2020 International Conference on Information and Communication Technology Convergence*, 2020
- [17] O.A. Malik, N. Ismail, B.R. Hussein, U. Yahya, “Automated real-time identification of medicinal plants species in natural environment using deep learning models-A case study from borneo region”, *Plants* 11 (2022)
- [18] A.M. Raghukumar, G. Narayanan, “Comparison of machine learning algorithms for detection of medicinal plants”, in: *Proceedings of the 4th International Conference on Computing Methodologies and Communication*, vol. 2020, 2020
- [19] Ghazi, M.M.; Yanikoglu, B.; Aptoula, E. “Plant identification using deep neural networks via optimization of transfer learning parameters”. *Neurocomputing* 2017
- [20] Reddy, S.R.; Varma, G.P.; Davuluri, R.L. “Optimized convolutional neural network model for plant species identification from leaf images using computer vision”. *Int. J. Speech Technol.* 2021
- [21] Aptoula E, Yanikoglu B (2013) “Morphological features for leaf-based plant recognition”. In *IEEE international conference on image processing*. IEEE, Melbourne
- [22] Bagal V, Manza R (2016) “Feature extraction of plant species from leaf architecture”. In: *International conference on electrical, electronics, and optimization techniques (ICEEOT)*. IEEE, Chennai
- [23] Chaki J, Parekh R, Bhattacharya S (2015) “Plant leaf recognition using texture and shape features with neural classifiers”. *Pattern Recognition Lett* 58:61–68.
- [24] Chothe S, Ratnaparkhe V (2015) “Plant identification using leaf images”. *Int J Innov Res Sci Eng Technol* 4(6):660–662.
- [25] Dahigaonkar T, Kalyane R (2018) “Identification of ayurvedic medicinal plants by image processing of leaf samples”. *Int Res J Eng Technol* 5(5):351–355.